

Research papers

Nonlinear control design and stability analysis of hybrid grid-connected photovoltaic-Battery energy storage system with ANN-MPPT method

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ABSTRACT

The problem of controlling a grid-connected solar energy conversion system with battery energy storage is addressed in this work. The study's target consists of a series and parallel combination of solar panel, DC/DC converter boost, DC/AC inverter, DC/DC converter buck-boost, Li-ion battery, and DC load. The main objectives of this work are: (i) PV voltage regulation: the PV panel voltage must track a given reference voltage corresponding to the maximum power point. The optimal reference voltage is generated by an optimizer based on the artificial neural network (ANN); (ii) DC link voltage regulation: The DC link voltage must track a reference voltage as closely as possible. (iii) PFC requirement: The grid currents must be sinusoidal with the same frequency and in phase with the voltage grid. The periodic nature of solar energy and the frequent fluctuations in load demand reduce battery life and charging performance. To deal with these limitations and ensure the battery's safety, a battery energy management algorithm is developed with the following objectives: (iv) CC mode: When the battery's state of charge is less than 100% ($SOC < 100\%$), it indicates that the battery is not fully charged, this aim has been adopted. The battery current should follow a constant reference as closely as possible to the permissible limits battery current; (v) CV mode: When the battery is completely charged, this mode is switched on. The voltage at the battery terminals then achieves its reference signal, which corresponds to the charge state's of the battery $SOC = 100\%$. In addition, the energy management system which generates several energy flow scenarios, in this work, the focus is to balance the energy flows between the load and the different energy sources to minimize the system costs, to ensure the stability of the grid and to improve the power quality. To do this, a mathematical modeling of the overall system was performed. Subsequently, backstepping controllers are then synthesized in order to ensure the control objectives. The closed loop control convergence is formally analyzed using Lyapunov's stability theory and its performances are illustrated by simulation. As a result, the simulation results indicate that the proposed controllers perform admirably in achieving their objectives.

1. Introduction

Solar energy has experienced unprecedented growth over the last few decades. In order to be part of the energy transition issue, major efforts are being made to develop electrical systems based on renewable resources [1,2]. The autonomous photovoltaic system is perfectly suited to a large number of applications such as the electrification of rural dwellings, on-board applications, telecommunications, etc. Indeed, the produced energy is generally used on site and without losses due to transport [3]. To connect photovoltaic systems to the electric grid, a DC/DC and a DC/AC converter are used [4]. To improve the power quality of the PV system, the design of a suitable controllers to drive the power electronic interface is one major issues [5,6].

In the literature different types of MPPT controls are presented [7–9]. Some of them are based on hill climbing techniques, of which the MPPT control tries to raise the operating point of the PV array along the characteristic ($V_{pv} - P_{pv}$) until reaching the maximum power point (MPP) [10]. However, the (P&O) control imposes oscillations around the optimal point. Indeed, this variation is necessary to know the variations of the PV output power and thus makes the operating point oscillates, as close as possible to the optimal point [11,12]. This results in steady state losses. On the other hand, the conductance increment method is based on the derivative of the conductance of the PV array to know the relative position of the MPP and finally apply an appropriate control action to the monitoring of this MPP. These two

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Nomenclature

V_{dc}	DC link voltage
i_{Li}	Current in inductor L_i
i_{bb}	Buck-boost input current
i_{L1}	Current in inductor L_1
R_s	Battery internal resistance
$R_p C_p$	Battery long term transient polarization effect
i_L	Load current
P_L	Load power
LF	Lyapunov function
e_{gd}, e_{gq}	dq -grid voltages components
i_{gd}, i_{gq}	Grid currents in dq -coordinates
ω_g	Grid pulsation
μ	Average values of D over cutting period PWM
i_b	Battery current
P_b	Battery power
U_{ocv}	Battery open circuit voltage
$MPPT$	Maximum Power Point Tracking
SOC	State of Charge
PFM	Power flow management
CC	Constant Current
ACC	Adaptive Constant Current
CV	Constant Voltage
PWM	Pulse Width Modulation
x	Average state
D	Duty ratio function

methods, although theoretically attractive and easy to study, present on the other hand difficulties of implementation, important noises and drifts difficult to filter and a need to take into account the dynamics of measurement which is generally neglected [13–15].

Recently, artificial neural network method has sparked renewed interest in photovoltaic systems. *ANN*'s ability to approximate unknown parameters inspired its use in *MPP* tracking.

The *ANN* (Artificial Neural Network) *MPPT* controller is suggested in this study to generate the reference voltage that meets the *MPPT* requirement under various atmospheric conditions. In practice, a trained neural network can quickly map the nonlinear relationship between input data and output data. Experiment data is used to create the *ANN* model. There are two modes of development for the neural network optimizer. Offline mode is required for testing a variety of neural network parameters in order to find the best neural network controller (structure, activation function, and training algorithm). Furthermore, in the energy conversion system, the On-line mode with the best *ANN* optimizer is used. The suggested neural network *MPPT* controller is evaluated and verified for various conditions using the Matlab/Simulink model.

The primary operating necessity of energy storage systems (*ESS*), when used in grid-connected *PV* systems, is to optimize power flow, typically in terms of energy costs [16]. Based on the condition of the *PV* system, grid-tied or isolated. A battery energy storage system (*BESS*) is a flexible device that can absorb and transmit power during different periods, becoming a possible solution to oppose and decrease the fluctuations of the output power of *PV* arrays [17–19]. There is no doubt that the supplemental investment and operational costs of *BESS* will affect the profitability of photovoltaic systems [20–22].

In the literature, numerous research has been devoted to the development of the *BESS* with a single energy storage devices with

the goal of either smoothing the *PV* output power effect or improving the *BESS*'s economic efficiency [23–25]. Various battery types have various performance characteristics, such as capital and operating costs, capacity degradation, battery life, and the limits of battery state of charge (*SOC*) [26,27]. From the technical consideration of energy storage devices, the performance of *BESS* can be improved by integrating different energy storage technologies. The lithium-ion battery based on *BESS* has seen widespread use in the systems of energy storage, particularly in decentralized systems for generating power. In comparison to nickel-cadmium and lead-acid batteries [28]. The Li-ion battery has a high specific energy, power density, a high cell voltage, and no memory effect, is non-polluting, has a long life, and has a low self discharge rate [29].

Power electronics are at the heart of the *PV* system and can have very important impacts on the yield, reliability and quality of the energy produced. Grid-connected *PV* systems are highly nonlinear, due to the diode current of the *PV* cell, the switching functions of the converters, the inverters and the energy storage system in case of use, requiring the design of high performance nonlinear controllers. In order to analyze and ensure the stability of grid-connected photovoltaic systems with *BESS*, in the literature, several nonlinear control techniques are proposed, such as feedback linearization, predictive model and sliding mode control. For the sake of simplicity, Table 1 summarizes the various approaches to controlling power electronics in a hybrid grid-connected photovoltaic-battery energy storage system.

Power management in hybrid systems based on renewable energies and storage systems plays a crucial role in ensuring efficient utilization of resources and reliable energy supply. The integration of renewable energy sources, such as solar, with energy storage systems presents opportunities to address the intermittent nature of these sources and enhance overall system performance [41]. Effective power management strategies are essential to optimize the utilization of renewable energy and storage, considering factors such as energy demand, weather conditions, and storage capacity [42,43]. Various techniques have been proposed for energy management, including model predictive control, fuzzy logic, and neural networks. These approaches enable intelligent decision-making, dynamic load balancing, and energy arbitrage [44,45]. However, due to their inherent nature, these methods remain complex and imprecise and may involve a large number of tuning parameters. In this paper, an energy flow management algorithm is employed to develop a robust, adaptive energy management strategy and optimal utilization of available resources, ensuring system stability, reliability, and cost-effectiveness.

In this paper, the nonlinear controllers are developed using the backstepping technique. These new controllers ensure that, firstly, the module voltage tracks the reference signal generated by the artificial neural network-based optimizer, secondly, the *DC* link voltage must follow its reference voltage as closely as possible, and thirdly, the grid currents always remain in phase and the same frequency with the grid voltage, according to the *PFC* requirement, an algorithm has been developed that allows switching, depending on the battery state of charge, between the mode where the battery current is forced to track its desired constant reference and the mode where the battery voltage is maintained equal to its constant reference.

The remainder of the article is structured as follows: Section 2 is dedicated to the description and modeling of the studied system; Section 3 discusses The controller design using the backstepping method; Section 4 describes the power flow management system (*PFM*) algorithm. Section 5 employs numerical simulations to achieve theoretical results. Finally, a conclusion and a reference list end the paper.

2. System modelling

The studied energy conversion system is shown in Fig. 1. It consist of, on one side, the *PV* panels and storage battery system, and on the other side, an inverter that is connected to the grid and a *DC* load.

Table 1
Power electronics in a hybrid grid-connected photovoltaic-battery energy storage system.

Control & management approaches	Power Converter & BESS Types	Objectives	Contributions	Drawbacks
Model predictive strategy [30,31]	Battery and converters	Cost minimization	power loss reduction	Difficult to implement in real time
Fuzzy logic control [32–34]	Boost/buck-boost converter	Control optimization	generation of optimal control pulse	Introduction of additional complexity to the DC/DC; Converter making it harder to design and troubleshoot
Input-output feedback linearization [35–37]	Boost/DC-AC inverter/battery	Stable operation for a hybrid microgrid	reliability is improved	Saturation in power electronic can affect the control; Performance and lead to undesirable behavior in the system
Sliding Mode approach [38–40]	DC/DC converter/battery	Energy management	Voltage regulation and quality of current injected to loads	High implementation complexity due to the need for accurate system modeling

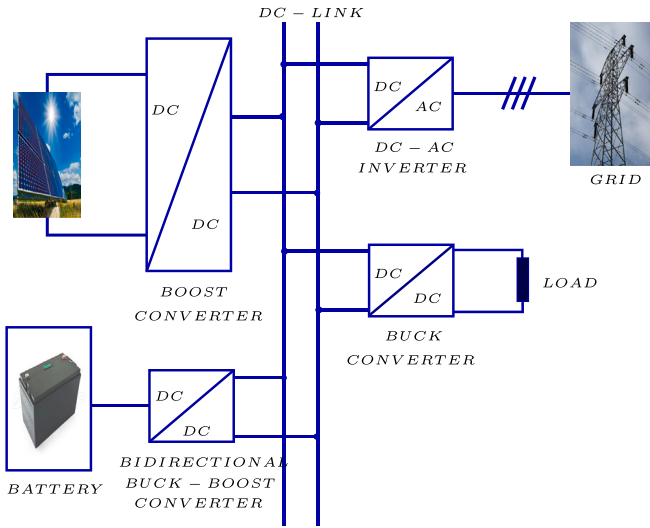


Fig. 1. The proposed grid-connected PV system with a battery energy storage system.

The DC/DC-boost converter is placed between the PV panels and the DC-link to allow the generated power to supply the loads directly and send the excess power to the storage system and/or the grid whenever the extracted power exceeds the power required by the loads. When the power demand is higher than the production of the PV system, additional energy will be generated either by the battery bank via the buck-boost converter or by the AC-grid via the inverter.

2.1. Photovoltaic module modeling

A photovoltaic module is an instantaneous converter that can only supply electrical power if it receives solar radiation. A cell in total darkness behaves as a passive component. The solar cell cannot be compared to any other conventional DC generator. It is neither a constant voltage source nor a constant current source. It has non-linear electrical characteristics depending on the solar radiation. The current (power) generated by PV systems is a function of several variables. Mainly the solar irradiation, the temperature, the efficiency of the PV panels, size of the PV panels and other parameters related to the manufacturing of the panels [46]. In the literature, the model of the PV cell is presented by an equivalent diagram Fig. 2(a). In addition to a current source in parallel with a diode, there are two resistors

describing the losses: one in series R_s , and the other in shunt denoted as R_{sh} . The current expression of the photocell is:

$$I = I_{ph} - I_d - I_{sh} \quad (1)$$

The mathematical equations representing the output current (I_{pv}) of the PV module can be expressed as follows [11,47]

$$\begin{aligned}
 I_{ph} &= [I_{sc} + K_i(T - 298)] \frac{G}{1000} \\
 I_d &= I_0 \left[\exp \left(\frac{q(V + R_s I)}{nKT} \right) - 1 \right] \\
 I_{rs} &= \frac{I_{sc}}{\left(\exp \left(\frac{qV_{oc}}{nN_s KT} \right) - 1 \right)} \\
 I_{sh} &= \left(\frac{V + R_s I}{R_{sh}} \right) \\
 I_0 &= I_{rs} \left(\frac{T}{T_n} \right)^3 \exp \left(- \frac{qE_{g0} \left(\frac{1}{T_n} - \frac{1}{T} \right)}{nK} \right)
 \end{aligned} \quad (2)$$

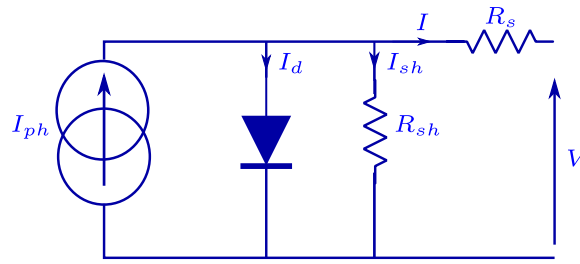
Where I_{ph} is the cell photocurrent, which is proportional to the irradiation G . This current also corresponds to the short-circuit current I_{sc} ; I_d denotes the current flowing through the diode; I_{rs} denotes the diode's reverse saturation current. I_{sh} is the current flowing through the shunt resistor. I_0 is the saturation current. I_{sc} stands for current short circuit; K_i is the short-circuit current of a cell at 25 °C and 1000 W/m²; T is the operating temperature; T_n is the nominal temperature, which is 298 °K; G is the solar irradiation; and q is the electron charge. V_{oc} stands for open circuit voltage. In reality, n is the ideality factor of the solar cell, which ranges between 1 and 5. The Boltzmann constant is K , and the semiconductor gap energy is E_{g0} ; N_s denotes the number of series cells; R_s denotes the series resistor; and R_{sh} denotes the shunt resistor. From these different equations, we can derive the implicit expression for the current delivered by a photovoltaic cell:

$$I = I_{ph} - I_0 \left[\exp \left(\frac{q(V + R_s I)}{nKT} \right) - 1 \right] - I_{sh} \quad (3)$$

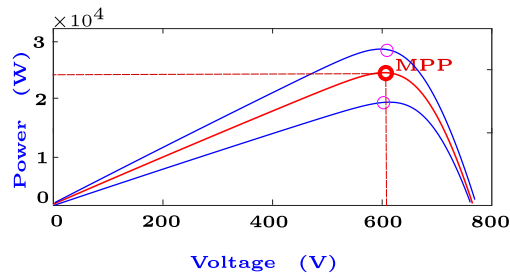
Where V is the photovoltaic cell voltage and I is the photovoltaic cell current [48,49]. This study is valid for all types and manufacturers of photovoltaic panels. We have chosen the (1Soltech 1STH-FRL-4H-245-M60-BLK) type from the company (Soltech) because its model is provided by (Matlab Simpower Toolbox). It consists of 20 series and 5 parallel connected modules. The module's electrical characteristics are listed in Table 3.

2.2. DC/DC boost converter and DC/AC inverter modeling

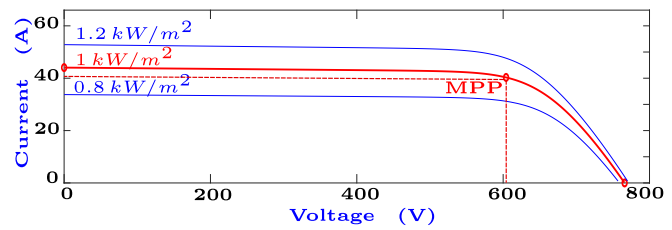
A configuration of a three-phase grid connected photovoltaic system is illustrated by Fig. 3. It consists of a PV array, an input capacitor



(a) Equivalent diagram of a photovoltaic cell.



(b) Power/voltage of photovoltaic module.



(c) Current/voltage of photovoltaic module.

Fig. 2. PV characteristics.

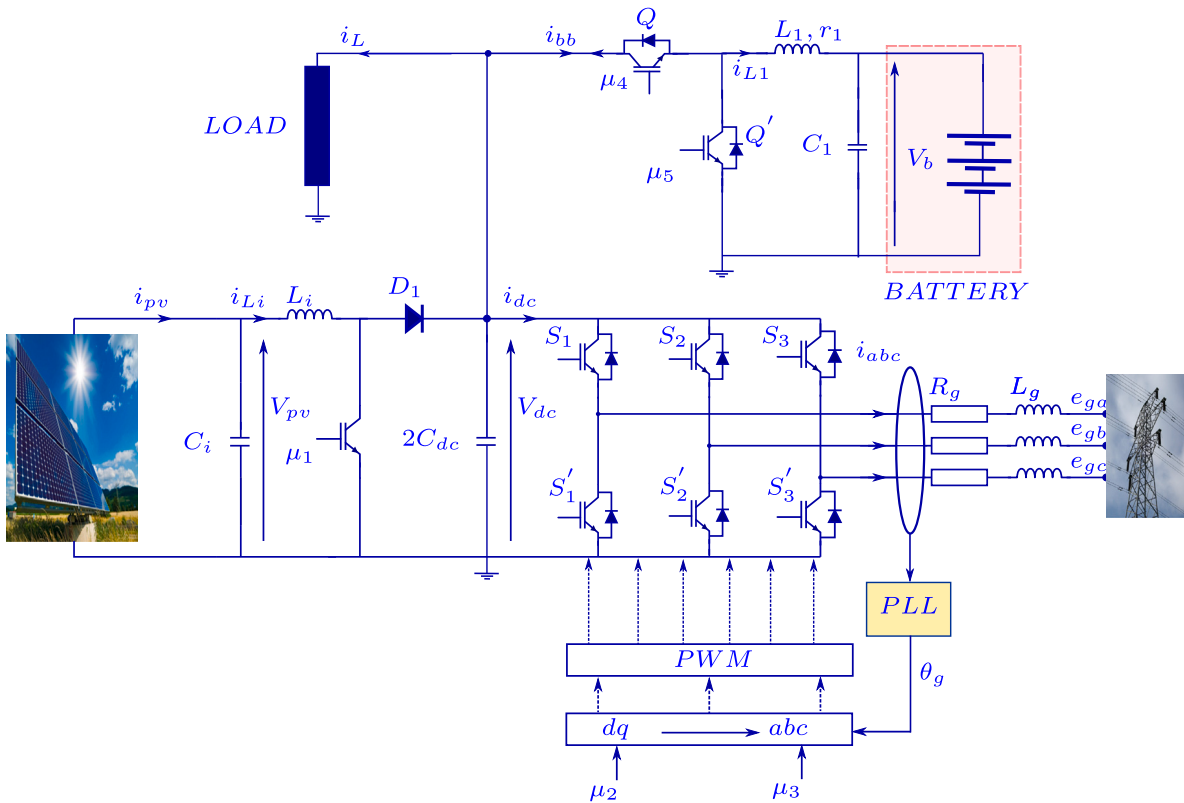


Fig. 3. Schematic diagram of grid-connected PV system.

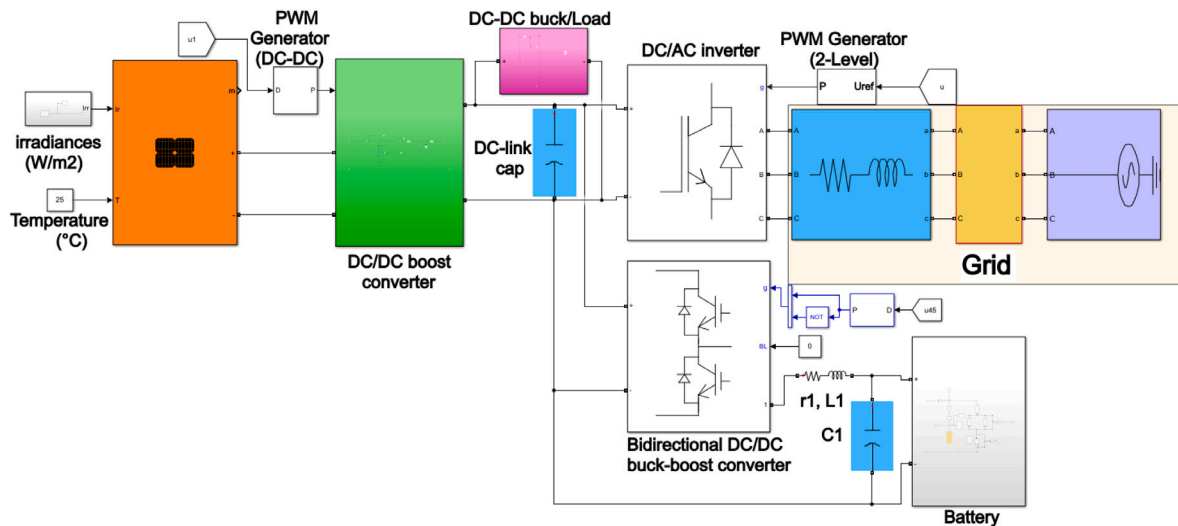


Fig. 4. MATLAB/Simulink model of grid-connected PV system.

C_i , a DC/DC converter to boost the PV voltage, a DC link capacitor C_{dc} , the three-phase bridge inverter, which consists of six IGBT's with anti-parallel diodes, is connected to the grid. Applying Kirchhoff's laws, this subsystem ($PV - DC/DC - DC/AC$ converters) is described by the following equations [50,51]:

$$\begin{aligned}
\frac{dV_{pv}}{dt} &= \frac{1}{C_i}(i_{pv} - i_{Li}) \\
\frac{di_{Li}}{dt} &= \frac{V_{pv}}{L_i} - \frac{(1 - \mu_1)}{L_i}V_{dc} \\
\frac{dV_{dc}^2}{dt} &= (1 - \mu_1)\frac{i_{Li}V_{dc}}{C_{dc}} - \frac{E_{gd}}{C_{dc}}i_{gd} - \frac{E_{gq}}{C_{dc}}i_{gq} - \frac{(P_b + P_L)}{C_{dc}} \\
\frac{di_{gd}}{dt} &= -\frac{R_g}{L_g}i_{gd} + \omega_g i_{gq} - \frac{E_{gd}}{L_g} + \frac{V_{dc}}{L_g}\mu_2 \\
\frac{di_{gq}}{dt} &= -\frac{R_g}{L_g}i_{gq} - \omega_g i_{gd} - \frac{E_{gq}}{L_g} + \frac{V_{dc}}{L_g}\mu_3
\end{aligned} \tag{4}$$

Where V_{pv} is the PV voltage, i_{pv} is the PV current, i_{L_i} is the Current in inductor L_i , V_{dc} is a DC link voltage, P_b and P_L are the powers of the battery and the load respectively. $[i_{abc}] = [i_a \ i_b \ i_c]^T$ is the grid currents, $[e_{gabc}] = [e_{ga} \ e_{gb} \ e_{gc}]^T$ is the sinusoidal triphase net voltages, i_{dc} designates the input current inverter, and S_i is the switch position function taking values in the discrete set $\{1, 0\}$. Specifically: $S_i = \begin{cases} 1 & \text{if } S_i \text{ ON and } S'_i \text{ OFF} \\ 0 & \text{if } S_i \text{ OFF and } S'_i \text{ ON} \end{cases} \ i = 1, 2, 3$ μ_2 and μ_3 are (averaged versions) of the dq -components of the triphase duty ratio system (S_1, S_2, S_3). Specifically, these inputs are obtained by operating the Park transformation on the latter and by averaging the result over the PWM periods (see Fig. 4).

2.3. Bidirectional DC – DC power converter and battery modeling

The aim, now is to elaborate a model of the half-bridge bidirectional *DC/DC* power converter and the battery, illustrated in Fig. 6. In this paper, we look at the battery electrical model, which consists of a parallel *RC* circuit connected in series with the battery's internal resistor [52,53].

In fact, Fig. 5(a) depicts the resulting equivalent circuit model of the battery, where R_s is the ohmic internal resistance that characterizes the battery energy losses; the first order $R_p C_p$ circuit denotes the long-term transient battery polarization effect; i_b is the battery current; and U_{ocv} is the open circuit voltage, which varies nonlinearly with the battery

state of charge ($U_{ocv} = f(SOC)$), as given in Fig. 5(b). Before we begin modeling the subsystem, let us first define the switching variable “ S ” which describes the bidirectional DC/DC power converter’s two operating modes:

$$S = \begin{cases} 1 & \text{during (Buck mode)} \\ 0 & \text{during (Boost mode)} \end{cases}$$

This power converter consists of two switches “ Q ” and “ Q' ”, a resistor r_1 in series with an inductance L_1 , a filtering capacitor C_1 followed by a battery energy storage system. This power converter is mainly used to interface the battery with the DC -link. It can operate as a buck or a boost converter depending on the energy flow direction. The buck mode ($S = 1$), ensures an adequate battery charging, shown in Fig. 6(a), the switch “ Q ” is controlled by a PWM signal ($\mu_4 \in \{0, 1\}$) while the switch “ Q' ” remains open ($\mu_5 = 0$). The boost mode ($S = 0$), guarantees the safety battery discharging, shown in Fig. 6(b), the switch “ Q' ” is controlled by a PWM signal ($\mu_5 \in \{0, 1\}$) while the switch “ Q ” remains open ($\mu_4 = 0$). According to the electric circuit, shown in Fig. 6, and considering the two operating modes described previously, applying Kirchhoff's laws, the instantaneous model of the bidirectional DC/DC power converter connected to the battery is as follows:

$$\begin{aligned}\frac{di_{L1}}{dt} &= -\frac{r_1}{L_1}i_{L1} - \frac{V_b}{L_1} + \mu_{45}\frac{V_{dc}}{L_1} \\ \frac{dV_b}{dt} &= \frac{i_{L1}}{C_1} - \frac{V_b}{R_s C_1} + \frac{V_p}{R_s C_1} + \frac{U_{ocv}}{R_s C_1} \\ \frac{dV_p}{dt} &= \frac{V_b}{R_s C_p} - \left(\frac{1}{R_s C_p} + \frac{1}{R_p C_p}\right)V_p - \frac{U_{ocv}}{R_s C_p}\end{aligned}\quad (5)$$

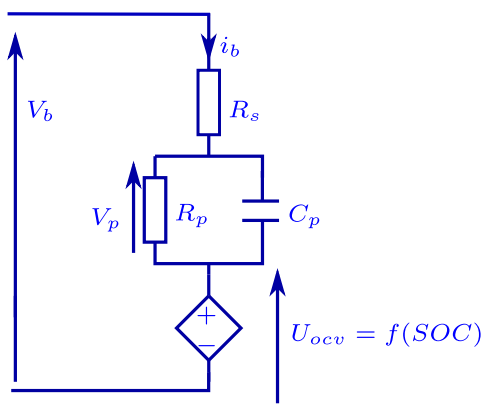
Where V_b denotes the battery voltage; i_{L1} is the inductor current L_1 ; V_p is the polarization voltage of the capacitor C_p . The buck boost control signal μ_{45} and input current i_{bb} can be obtained respectively as follows (see Fig. 7):

$$\mu_{45} = S\mu_4 + (1 - S)(1 - \mu_5) \quad (6)$$

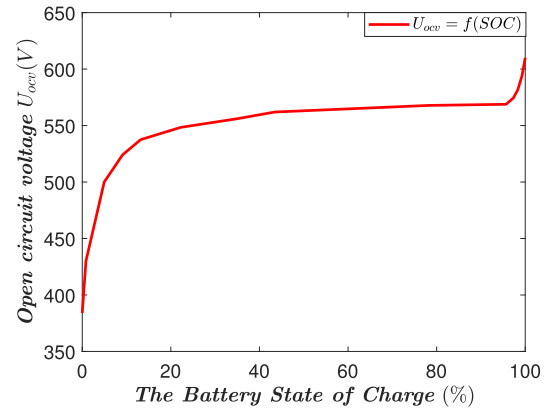
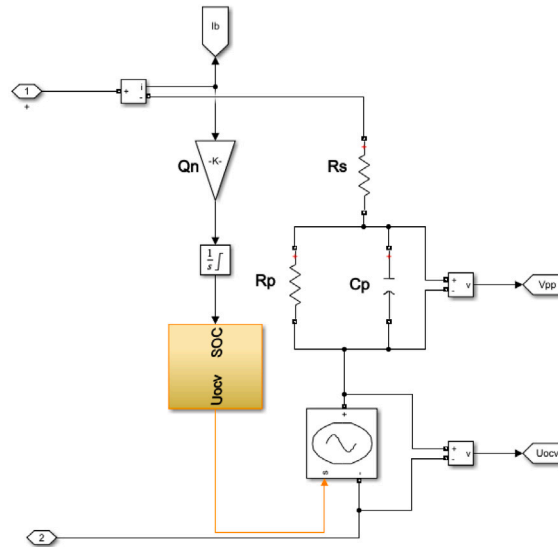
$$i_{bb} = (S\mu_4 + (1 - S)(1 - \mu_5))i_{L1} = \mu_{45}i_{L1} \quad (7)$$

2.4. Overall system model

Based on the previous subsystem models (4), (5) and (7), the aim now is to determine the model of the whole system that is useful for control design purposes. By combining (4) and (5), and for more convenience, we consider the following averaged model for control



(a) Model of the battery equivalent circuit.

(b) The battery open circuit voltage U_{ocv} .

(c) MATLAB/Simulink model of battery.

Fig. 5. Equivalent electrical circuit of Lithium-ion battery.

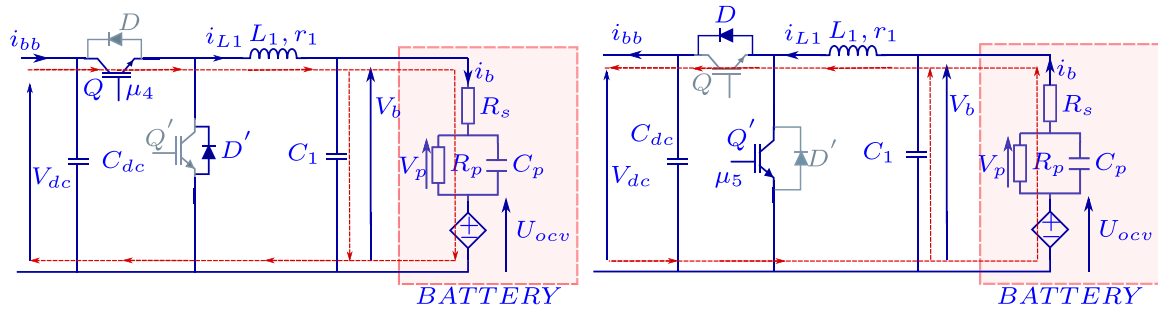
(a) Buck mode ($S = 1$, charging mode).(b) Boost mode ($S = 0$, discharging mode).

Fig. 6. Bidirectional Buck-Boost Converter. (a) Buck Operating Mode: Charging Mode; (b) Boost Operating Mode: Discharging Mode.

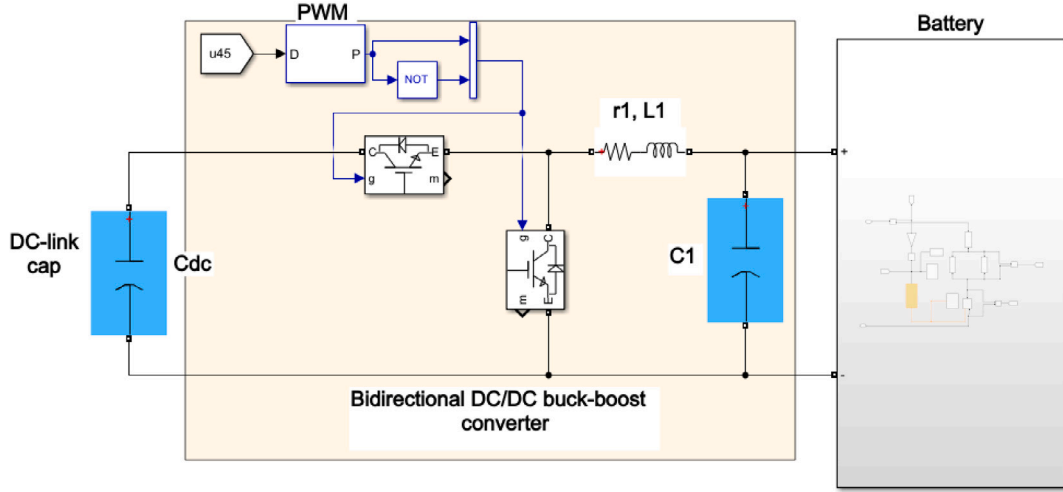


Fig. 7. MATLAB/Simulink model of bidirectional *DC/DC* buck-boost Converter with battery.

design purposes. This model is obtained by averaging the instantaneous model over the *PWM* cutting period.

$$\dot{x}_1 = \frac{1}{C_i}(i_{pv} - x_2) \quad (8a)$$

$$\dot{x}_2 = \frac{x_1}{L_i} - \frac{(1 - \mu_1)}{L_i} \sqrt{x_3} \quad (8b)$$

$$\dot{x}_3 = (1 - \mu_1) \frac{x_2 \sqrt{x_3}}{C_{dc}} - \frac{E_{gd}}{C_{dc}} x_4 - \frac{E_{gq}}{C_{dc}} x_5 - \frac{(P_b + P_L)}{C_{dc}} \quad (8c)$$

$$\dot{x}_4 = -\frac{R_g}{L_g} x_4 + \omega_g x_5 - \frac{E_{gd}}{L_g} + \frac{\sqrt{x_3}}{L_g} \mu_2 \quad (8d)$$

$$\dot{x}_5 = -\frac{R_g}{L_g} x_5 - \omega_g x_4 - \frac{E_{gq}}{L_g} + \frac{\sqrt{x_3}}{L_g} \mu_3 \quad (8e)$$

$$\dot{x}_6 = -\frac{r_1}{L_1} x_6 - \frac{x_7}{L_1} + \mu_{45} \frac{\sqrt{x_3}}{L_1} \quad (8f)$$

$$\dot{x}_7 = \frac{x_6}{C_1} - ax_7 + ax_8 + aU_{ocv} \quad (8g)$$

$$\dot{x}_8 = bx_7 - cx_8 - bU_{ocv} \quad (8h)$$

Where $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ and μ denote respectively the average values of $V_{pv}, i_{L_i}, V_{dc}^2, i_{gd}, i_{gq}, i_{L_1}, V_b, V_p$ and D over the switching period; The remaining notation is defined as follows:

$$a = \frac{1}{R_s C_1}; b = \frac{1}{R_s C_p}; c = \left(\frac{1}{R_s C_1} + \frac{1}{R_s C_p} \right)$$

3. Nonlinear controller design

3.1. Control objectives

Based on the model (8a)–(8h) developed previously, the aim now is to design a nonlinear controller. There are five operational control objectives:

(i) **PV voltage regulation:** The voltage V_{pv} must track a given reference voltage that corresponds to the maximum power point (Fig. 2(b)). The optimal reference voltage V_{opt} is generated by an optimizer based on an artificial neural network.

(ii) **DC link voltage regulation:** The *DC*-link voltage V_{dc} must track a reference voltage V_{dref} as accurately as possible.

(iii) **PFC requirement:** The line grid currents (i_a, i_b, i_c) must be sinusoidal with the same frequency and in phase with the grid voltages (e_{ga}, e_{gb}, e_{gc}).

The last two control objectives are constantly modified based on the battery state of charge (*SOC*):

(iv) **CC mode:** When the battery *SOC* < 100%, (i.e., the battery is not fully charged), the battery current i_b must track a constant reference as closely as possible, which corresponds to the maximum permissible battery current denoted as i_{bref} , (30 A in this work).

(v) **CV mode:** When the battery is fully charged, this mode is activated. The battery voltage must be regulated to a constant value V_{bref} , which corresponds to *SOC* = 100%.

3.2. DC/DC boost converter control design

3.2.1. The reference voltage optimizer technique

The artificial neural network (*ANN*) has recently seen significant advancements not only in theory but also in application [54,55]. Fig. 8 shows a common artificial neural network with several layers: input layers, hidden layers, and output layers. *PV* array parameters such as voltages and currents, environmental data such as irradiance and temperature, or any combined effect of these can be used as input to the *ANN* for *MPPT*. Typically, the output is a reference signal that forces the *PV* array to operate at or near the *MPP*. The input and output data are obtained through experimental measurements. This technique involves two stages to track the *MPP* of a *PV* array. In the first stage, the trained *ANN* guides the reference signals (V_{ref}, P_{ref}) to the optimal values (V_{opt}, P_{opt}), which are nearly equal to the *MPP*. The training data are generated by simulating the *PV* array using its parameters in the Matlab/Simulink/SimPowerSystems environment. The power-voltage characteristic for the *ANN* input data is recorded for each case, considering specific irradiance and temperature values. The *MPP* of this characteristic is also recorded for the *ANN* output data. The neural network is implemented to determine the voltage of the *PV* array with the highest power. The network is created by training it in a supervised manner using the *trainlm* function, which employs the Levenberg–Marquardt algorithm. An acceptable training squared error of 2.68×10^{-7} is achieved. The first layer of the network has 10 logsig neurons, and the second layer has 1 purelin neuron. As a result, the total number of neurons in the network is 11, [56].

3.2.2. Controller design

Recall the control objective, which is to make the *PV* panel voltage follow the reference voltage V_{opt} given by the *ANN*-based optimizer as closely as possible. To achieve this objective, a control law that generates the control signal using the backstepping approach [57] will be employed. Let us define control errors as: $z_1 := x_1 - V_{opt}$ and

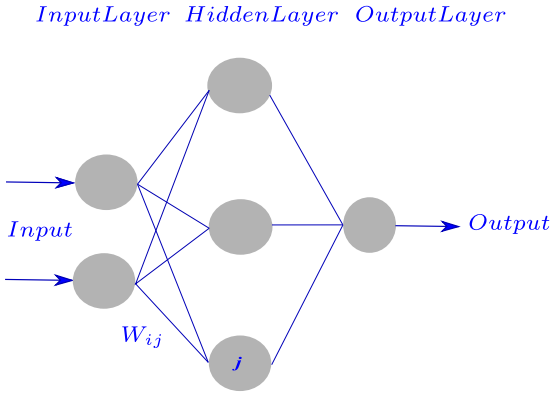


Fig. 8. ANN Diagram.

$z_2 := x_2 - x_2^*$. The variable (x_2^*) is a virtual input control that will be defined in the Appendix.

Proposition 1. Consider the subsystem described by state space equations (8a)–(8b) together with the control input:

$$\mu_1 = 1 - \frac{L_i}{\sqrt{x_3}} \left[\frac{x_1}{L_i} + c_2 z_2 - \dot{x}_2^* \right] \quad (9)$$

Where c_1 and c_2 are positives design parameters. The resulting closed loop system expressed in (z_1, z_2) coordinates is governed by the following equations:

$$\begin{aligned} \dot{z}_1 &= -c_1 z_1 + z_2 \\ \dot{z}_2 &= -c_2 z_2 - \dot{x}_2^* \end{aligned} \quad (10)$$

Then the resulting closed loop linear system (10) is globally asymptotically stable whatever the initial conditions.

Proof of Proposition 1. The stability analysis and proof of Proposition 1 are detailed in Appendix.

3.3. DC/AC inverter control design

To achieve these objectives, two control loops are required. The first one ensures the tracking of the DC voltage x_3 to a constant reference value x_3^* . The second loop concerns the injection of the reactive power $Q_g = E_{gq}x_4 - E_{gd}x_5$ which must be regulated to a reference signal denoted as Q_{gref} . Based on (8c), (8d) and (8e) where μ_2 and μ_3 are the actual input signals, the backstepping approach is performed in two steps. Let us define control errors as: $z_3 := x_3 - x_3^*$, $z_4 := \theta - \theta^*$, and $z_5 := x_5 - x_5^*$. The variables θ and virtual input control (θ^*) will be defined in Appendix. The following proposition summarizes the controller's proposed synthesis.

Proposition 2. Consider the control subsystem, consisting of the PV panel, a DC/DC converter and a DC/AC inverter connected to the grid, described by the state space equations (8c)–(8e) and the control input is given by:

$$\begin{bmatrix} \mu_2 \\ \mu_3 \end{bmatrix} = \frac{L_g C_{dc}}{\sqrt{x_3} E_g^2} \begin{bmatrix} E_{gd} & \frac{E_{gq}}{C_{dc}} \\ E_{gq} & -\frac{E_{gd}}{C_{dc}} \end{bmatrix}^{-1} \begin{bmatrix} -(c_3^2 - 1)z_3 + (c_3 + c_4)z_4 + g(x, t) \\ -c_5 z_5 - h(x, t) \end{bmatrix} \quad (11)$$

where c_3 , c_4 , and c_5 are positive design parameters. Then one has the following properties:

1. The resulting closed-loop sub-system is governed by the following equations:

$$\begin{aligned} \dot{z}_3 &= -c_3 z_3 + z_4 \\ \dot{z}_4 &= -c_4 z_4 - z_3 \\ \dot{z}_5 &= -c_5 z_5 \end{aligned} \quad (12)$$

2. The resulting closed-loop linear system (12) is GAS whatever the initial conditions, then the DC link voltage and PFC requirement are asymptotically fulfilled.

Proof of Proposition 2. The stability analysis and proof of Proposition 2 are detailed in Appendix A.2.

3.4. DC/DC buck-boost converter control design

The control design of the DC/DC buck-boost converter will be conducted to ensure the charging and discharging of the battery. These two modes are known as CC mode and CV mode, The CC mode is used when the DC/DC converter is set to buck mode, while the CV mode is used when the converter is set to boost mode.

3.4.1. CC mode controller design

When the battery's state of charge (SOC) is less than 100%, indicating that the battery is not fully charged, it is crucial for the battery current i_b to closely follow a constant reference value, representing the maximum permissible battery current denoted as i_{bref} .

This mode is applicable during daytime when the PV panel is operational. In this mode, it is necessary for the battery to remain not fully charged ($SOC < 100\%$), and the power generated by the PV panel should be greater than the sum of the battery's maximum power and the load power, regardless of the grid's state (ON or OFF). However, if the power generated by the PV panel is less than the sum of the battery's maximum power and the load power, the system switches to the adaptive Constant Current (ACC) mode. In both CC and ACC modes, the primary control objective is to regulate the battery current i_b to its desired constant value i_{bref} .

To achieve this control objective, the subsystem (8f)–(8h) is employed, and the control design is carried out in two steps using the backstepping approach. Before delving into the control design, we make the following assumption:

Assumption 1. Considering that the battery open-circuit voltage U_{ocv} changes at a much slower rate than the battery current dynamics, it is reasonable to assume that $\dot{U}_{ocv} \approx 0$.

Let us define control errors as: $z_6 := i_b - i_{bref}$ and $z_7 := ax_6 - \alpha^*$. The variable (α^*) is a virtual input control that will be defined in Appendix A.3.

Proposition 3. Consider the subsystem described by state space equations (8f)–(8h) and the following control input:

$$\begin{aligned} \mu_{45cc} = & \frac{L_1}{a\sqrt{x_3}} \left[(c_6^2 - 1)z_6 - (c_6 + c_7)z_7 + \left[\frac{ar_1}{L_1} + a(a+b) \right] x_6 \right. \\ & + \left[\frac{a}{L_1} - \frac{a(a+b) + b(a+c)}{R_s} \right] x_7 + \left[\frac{a(a+b) + c(a+c)}{R_s} \right] x_8 \\ & \left. + \left[\frac{a(a+b) + b(a+c)}{R_s} \right] U_{ocv} \right] \end{aligned} \quad (13)$$

where c_6 and c_7 are undefined positive constants. Then one has the following characteristics:

1. The resulting closed loop system expressed in (z_6, z_7) coordinates is governed by the following equations:

$$\begin{aligned} \dot{z}_6 &= -c_6 z_6 + z_7 \\ \dot{z}_7 &= -c_7 z_7 - z_6 \end{aligned} \quad (14)$$

2. The resulting closed loop linear system (14) is GAS whatever the initial conditions. As a result, for any positive design parameters c_6 and c_7 , the errors system (z_6, z_7) vanishes exponentially. In particular, the battery current i_b closely tracks its desired constant value i_{bref} as closely as possible.

Proof of Proposition 3. The stability analysis and proof of Proposition 3 are detailed in Appendix A.3

3.4.2. CV mode controller design

The control purposes in this subsection is to keep the battery voltage V_b constant and similar to its constant reference V_{bref} during the CV mode. Noted that this mode is activated when the battery is completely charged ($SOC = 100\%$) during the PV daytime mode whatever the state of grid, ON or OFF.

To achieve this control objective, the subsystem (8f)–(8h) is employed. Using the backstepping method, let us define control errors as: $z_8 := x_7 - V_{bref}$ and $z_9 := \frac{x_6}{C_1} - \gamma^*$. The variable (γ^*) is a virtual input control that will be defined in Appendix A.4.

Proposition 4. Consider the subsystem described by state space equations (8f)–(8h) and the following control input:

$$\mu_{45cv} = \frac{L_1 C_1}{\sqrt{x_3}} \left[(c_8^2 - 1)z_8 - (c_8 + c_9)z_9 + \left(\frac{a}{C_1} + \frac{r_1}{L_1 C_1} \right)x_6 + \left(\frac{1}{L_1 C_1} - a(a+b) \right)x_7 + a(a+c)x_8 + a(a+b)U_{ocv} \right] \quad (15)$$

where c_8 and c_9 are positives design parameters. Then,

1. the resulting closed loop sub-system (8f)–(8h) undergoes the following equations:

$$\begin{aligned} \dot{z}_8 &= -c_8 z_8 + z_9 \\ \dot{z}_9 &= -c_9 z_9 - z_8 \end{aligned} \quad (16)$$

2. The resulting closed loop linear system (16) is GAS. Therefore, the error vector (z_8, z_9) vanish exponentially, whatever the initial conditions. Then, the battery voltage V_b closely tracks its desired constant value V_{bref} as closely as possible.

Proof of Proposition 4. The stability analysis and proof of Proposition 4 are detailed in Appendix A.4

3.4.3. DC link controller design via DC/DC buck-boost converter

The DC link control with DC/DC buck-boost converter is adopted when the grid is disconnected during the daytime or at night, the V_{dc} voltage is regulated by the battery through the DC/DC buck-boost converter, therefore the Eq. (8c) is reduced as follows:

$$\dot{x}_3 = -\frac{P_b}{C_{dc}} - \frac{P_L}{C_{dc}} \quad (17)$$

We note that the battery power $P_b = V_b i_{L1} = x_6 x_7$, using the Eqs. (8f) and (8g), the dynamic of the battery power P_b undergoes the following equation:

$$\dot{P}_b = -\left(\frac{r_1}{L_1} + a\right)P_b + \mu_{45} \frac{x_7 \sqrt{x_3}}{L_1} + f(x, t) \quad (18)$$

With,

$$f(x, t) = -\frac{x_7^2}{L_1} + \frac{x_6^2}{C_1} + ax_6 x_8 + ax_6 U_{ocv} \quad (19)$$

Proposition 5. Consider the subsystem described by Eqs. (17)–(18) and the control input is given as follows:

$$\mu_{45dc} = \frac{L_1}{x_7 \sqrt{x_3}} \left[-c_{11} z_{11} + \left(\frac{r_1}{L_1} + a \right) P_b - f(x, t) + \dot{P}_b^* \right] \quad (20)$$

where c_{10} and c_{11} are positives design parameters. The resulting closed loop system expressed in (z_{10}, z_{11}) coordinates is governed by the following equations:

$$\begin{aligned} \dot{z}_{10} &= -c_{10} z_{10} + z_{11} \\ \dot{z}_{11} &= -c_{11} z_{11} - z_{10} \end{aligned} \quad (21)$$

Then the resulting closed loop linear system (21) is GAS whatever the initial conditions.

Proof of Proposition 5. The stability analysis and proof of Proposition 5 are detailed in Appendix A.5

3.5. DC link controller design via DC/DC boost converter

The DC link control with a DC/DC boost converter is employed when the grid is disconnected during the daytime. This mode is activated when the power generated by the photovoltaic panel exceeds the sum of the battery's maximum power (if the battery is not fully charged) and the required load power. In this situation, the system operates in the Adaptive Power Point Tracking (APPT) mode, and the DC link voltage V_{dc} is regulated to follow its reference value V_{dcref} . The boost DC/DC converter is utilized to ensure that the V_{dc} voltage remains at the desired level.

To achieve this control objective, let us define control error $z_{12} := x_3 - V_{dcref}$.

Proposition 6. Consider the subsystem described by Eq. (8c) and the control input is given by:

$$\mu_{1b} = 1 - \frac{C_{dc}}{x_2 \sqrt{x_3}} \left[-c_{12} z_{12} + \frac{E_{gd}}{C_{dc}} x_4 + \frac{E_{gq}}{C_{dc}} x_5 + \frac{(P_b + P_L)}{C_{dc}} + \dot{x}_3^* \right] \quad (22)$$

where c_{12} is positive design parameter. The resulting closed loop system expressed in (z_{10}) coordinates is governed by the following equations:

$$\dot{z}_{12} = -c_{12} z_{12} \quad (23)$$

Then the resulting closed loop linear system (23) is GAS whatever the initial conditions.

Proof of Proposition 6. The stability analysis and proof of Proposition 6 are detailed in Appendix A.6

After designing the controllers for the overall system, the block diagram for implementing the proposed controllers is given below Fig. 9:

4. Power flow management

Achieving global optimization of battery state of charge (SOC) involves implementing strategies and techniques to ensure the battery operates efficiently and effectively. Generally a BMS is used for optimizing battery SOC. Various parameters, including voltage, temperature, and current can be controlled. One of the aims of this work is to well-design a control algorithm to manage charging and discharging processes and ensure the battery operates within its optimal SOC range. Implementing advanced charge/discharge controller algorithm, depending of different factors such as battery chemistry, capacity, and balancing the power demands of connected loads and grid, to determine the optimal charging rate and duration. They prevent overcharging or deep discharging, which can degrade battery life and affect performance. The system can manage the battery's charging and discharging cycles for maximum efficiency.

Based on the available optimal PV power, battery state of charge, load power demand, and grid availability, several power flow management (PFM) scenarios are considered to balance the power flows between the load and different energy sources. These scenarios aim to minimize system costs (economic view), ensure grid stability, and improve power quality (technical view). To achieve those requirements of PFM, a flowchart is proposed in Figs. 10–12 to summarize the following operating modes.

Table 2
System characteristics.

Characteristics	Values	Characteristics	Values
PV array		DC/DC buck-boost	
Parallel strings	5	Inductor	$L_1 = 100 \text{ mH}$
Series-connected modules per string	20	resistor	$r_1 = 100 \text{ m}\Omega$
Three phase supply network	400 V/50 Hz	Capacitor	$C_1 = 100 \text{ mF}$
DC/DC boost		Modulation frequency	$F_m = 25 \text{ KHz}$
Capacitor	$C_i = 20 \mu\text{F}$	Li-ion battery	
Inductor	$L_i = 600 \text{ mH}$	Internal resistor	$R_s = 100 \text{ m}\Omega$
Modulation frequency	$F_m = 5 \text{ KHz}$	Polarization resistor	$R_p = 1 \mu\Omega$
DC Bus		Polarization capacitor	$C_p = 500 \text{ F}$
Capacitor	$C_{dc} = 75 \mu\text{F}$	Nominal voltage	525 V
DC/AC inverter		Maximal voltage	612 V
resistor	$R_s = 10 \text{ m}\Omega$	Nominal capacity	$Q_n = 50 \text{ Ah}$
Inductor	$L_s = 10 \text{ mH}$		
Modulation frequency	$F_m = 25 \text{ KHz}$		

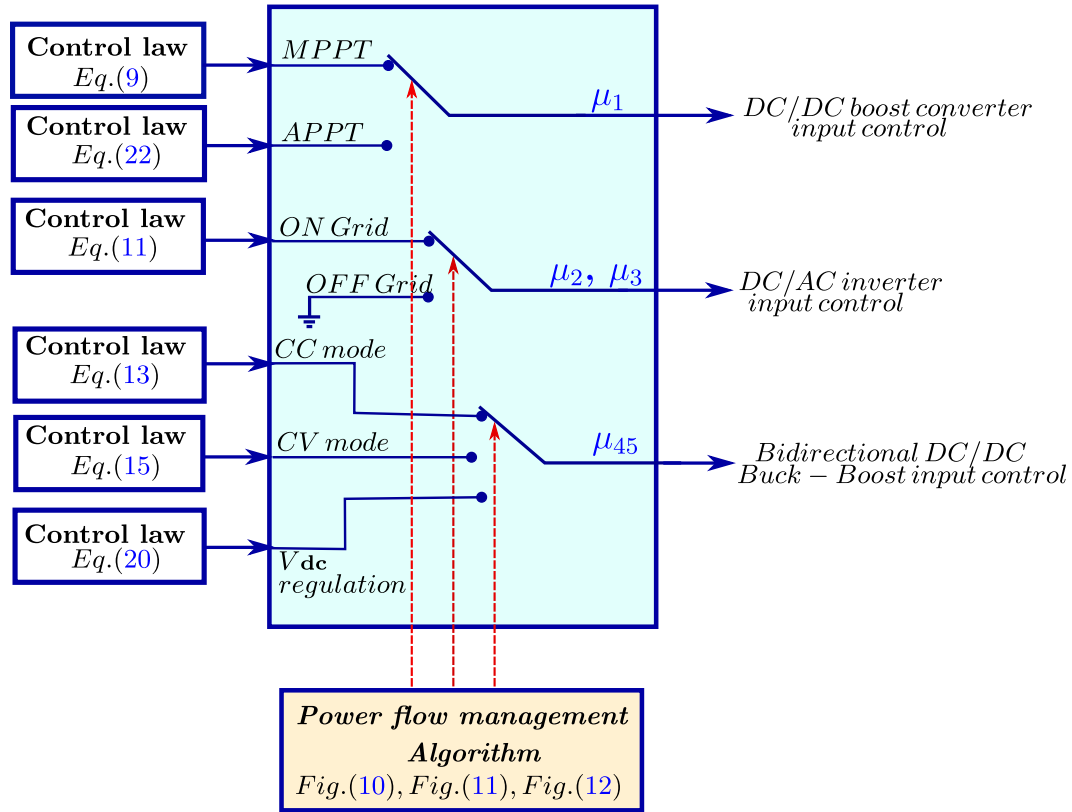


Fig. 9. Block diagram of the proposed controller implementation.

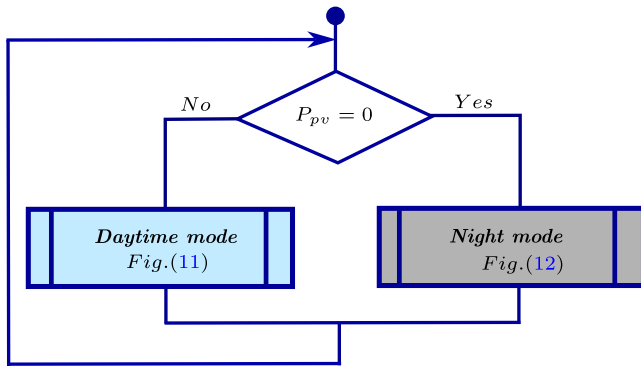


Fig. 10. The power flow management flowchart.

5. Simulation results

The experimental setup described by Fig. 13 has been simulated, within the Matlab/Simulink/SimPowerSystems environment with a calculation step of 50 μs , the System whose characteristics are summarized in Tables 2 and 3 (see Table 4).

5.1. Case 1. The power flow management system's performance in the daytime mode with grid OFF

During the interval time [0 8 s], the solar energy conversion with storage battery system is disconnected from the grid. The initial value of the battery SOC is 99.99% given by Fig. 14(b). The system is tested under step irradiance changes described at the top of Fig. 14(a). The available power P_{pv} and the optimal power P_{opt} at the output of the photovoltaic panel are shown at the bottom of Fig. 14(a). Over the interval time [0 3 s], the generated power by the photovoltaic panel is less than the battery's maximum power and the required load power,

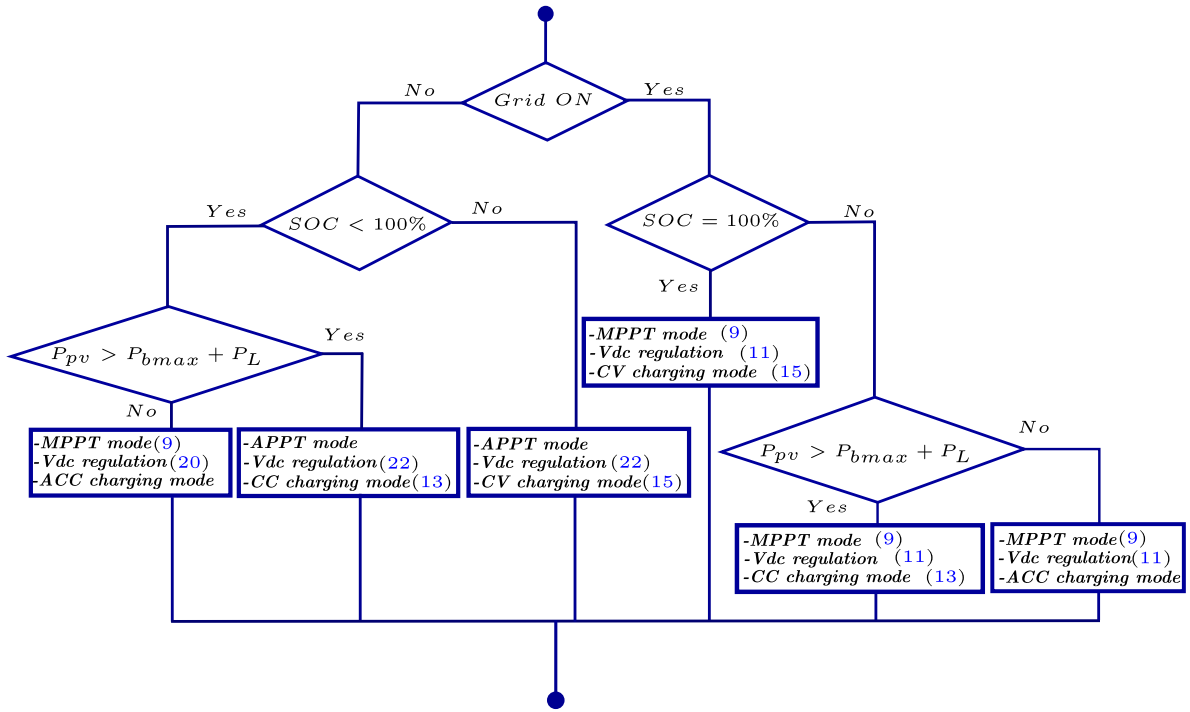


Fig. 11. The power flow management flowchart in the daytime mode.

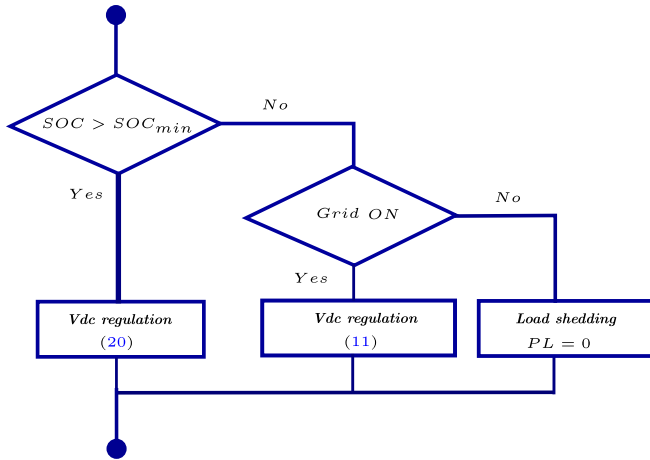


Fig. 12. The power flow management flowchart in the night mode.

as the battery SOC is less than 100% (Fig. 14(b)). The management algorithm switches the system to operate under $MPPT$ mode depicted in Fig. 14(a) with ACC mode as shown in Fig. 14(e). The desired battery current i_{bref} is given by the equation $i_{bref} = (P_{pv} - P_L)/V_b$. Moreover, the DC -link voltage V_{dc} perfectly tracks its reference V_{dcref} given by Fig. 14(d). The DC -link voltage regulation is ensured through the DC/DC buck-boost converter. During the interval time [3 s 5.7 s], the generated power by the photovoltaic panel becomes higher than the battery's maximum power plus the demanded load power. As seen in Fig. 14(b), the battery SOC is less than 100%. The management algorithm switches the system to operate under $APPT$ mode depicted in Fig. 14(a) and the CC mode (Fig. 14(e)). Additionally, the DC -link voltage V_{dc} follows its reference V_{dcref} as shown in Fig. 14(d). The V_{dc} voltage is ensured in this case by the boost DC/DC converter. Over the interval time [5.7 s 8 s], the battery is completely charged ($SOC = 100\%$) (Fig. 14(b)). However, the management algorithm switches the system to CV mode (Fig. 14(e)) to protect the battery from damage

Table 3

Electrical specifications of the solar module.

Characteristics	Values
Cells per module N_{cell}	60
Maximum power P_m (W)	243.412
Short circuit current I_{SC} (A)	8.81
Open circuit voltage V_{oc} (V)	38.3
Voltage at max power point V_{mp} (V)	30.2
Current at max power point I_{mp} (A)	8.06
Temperature coefficient of I_{sc} (%/deg C)	0.07
Temperature coefficient of V_{oc} (%/deg C)	-0.35601

Table 4

Controller design parameters.

Controller parameters	$c_1 = 5000$;	$c_2 = 5000$;	$c_3 = 700$;	$c_4 = 1200$;
	$c_5 = 6700$;	$c_6 = 100$;	$c_7 = 1000$;	$c_8 = 1000$;
	$c_9 = 1000$;	$c_{10} = 10$;	$c_{11} = 185$;	$c_{12} = 1000$.

or overheating, and the required power by the load is delivered by the photovoltaic panel in $APPT$ mode as shown in Fig. 14(f). In this case, the management algorithm controls the boost DC/DC converter to ensure the DC -link regulation. The overall control laws are described by Figs. 15(a)–15(b).

5.2. Case 2. The power flow management system's performance in the daytime mode with grid ON

During this case, the solar energy conversion with the battery storage system is connected to the grid, the irradiances profile is shown in Fig. 16(a), the management algorithm during this scenario always controls the DC/DC boost converter to operate in $MPPT$ mode as shown in Fig. 16(a). In addition, the DC -link voltage regulation and the PFC are ensured by the grid depicted in Figs. 16(c) and 16(f), over the interval time [0 3 s], the battery SOC is less than 100% and the generated power by the PV panel is less than the battery maximum power P_{max} and the load demanded power P_L as shown in Fig. 16(e), in this case, the management algorithm switches to ACC

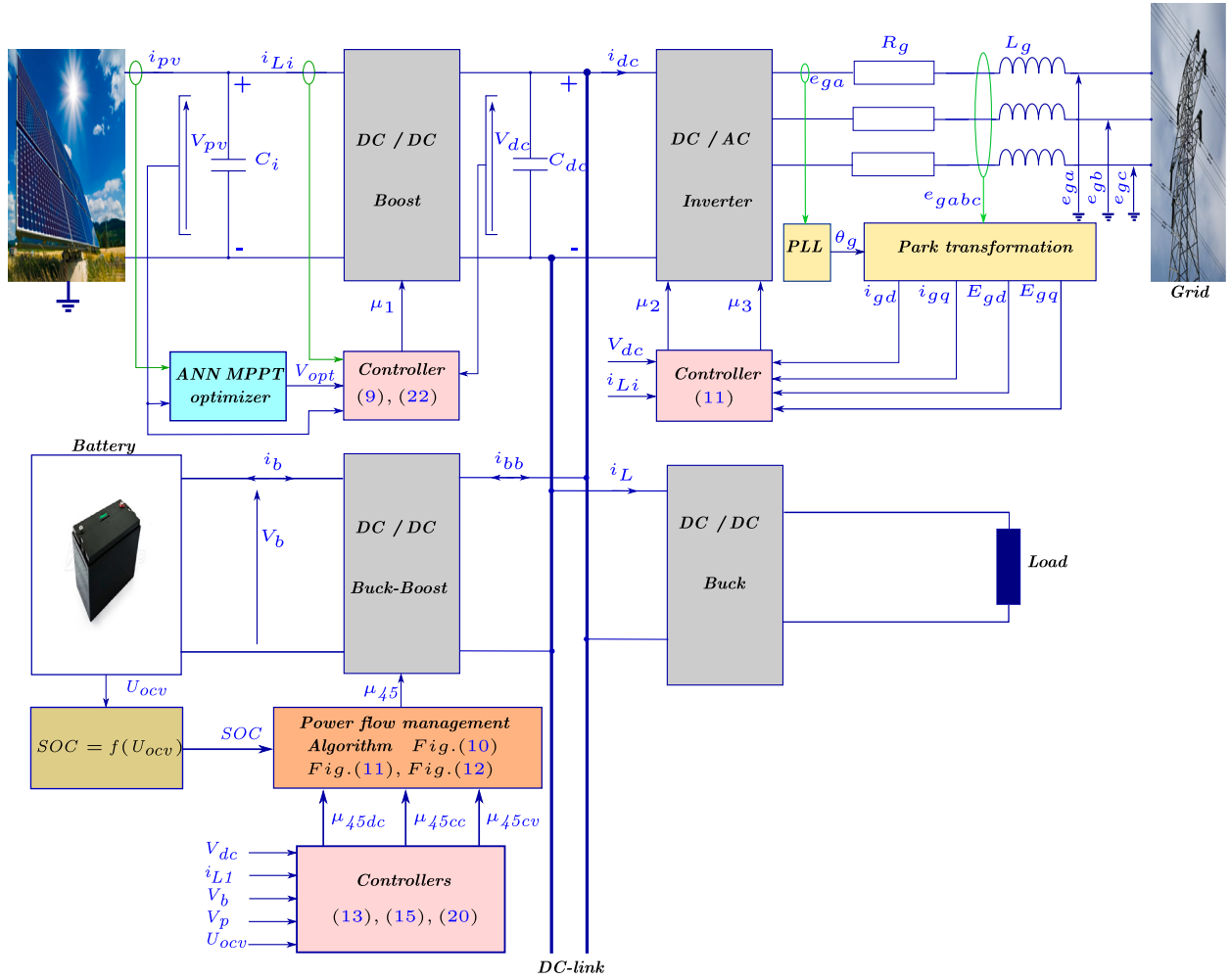


Fig. 13. Schema of the proposed control strategy.

mode with a current $i_{bref} = \frac{P_{pv}-P_L}{V_b}$ depicted in Fig. 16(d), during the interval time [3 s 5.7 s], the delivered power by the PV panel becomes higher than the battery maximum power plus the load power, since the battery SOC is less than 100%, the management algorithm switches to CC mode (Fig. 16(d)) and the excess power delivered by the PV panel is injected to the grid as shown in Fig. 16(e), over the interval time [5.7 s 8 s], the battery is completely charged $SOC = 100\%$ (Fig. 16(b)), the management algorithm controls the buck-boost DC/DC converter to operate in CV mode (Fig. 16(d)). Fig. 18 shows the grid current THD spectrum. The adopted control strategy has a positive effect on injected current quality. The obtained fundamental is about 47.36 A with frequency 50 Hz, and the obtained THD is 1.69%. This value is indeed much lower than the standard for low-voltage networks (typically below 5%). The overall control laws are shown in Figs. 17(a)–17(d).

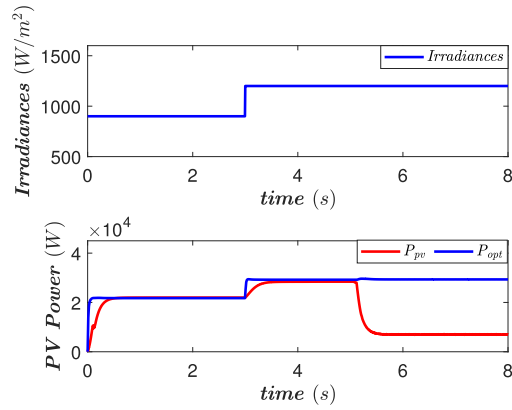
5.3. Case 3. The power flow management system's performance in the night mode

During the night, the delivered photovoltaic panel power is null, so we have to discuss some critical situations of energy management in the system, over the interval time [0 2.8 s], the battery SOC is higher than SOC_{min} , in this case, it is equal to 20% as described in Fig. 19(a), the demand load power is provided by the battery as indicated in Fig. 19(c). Moreover, the algorithm management controls the DC/DC buck-boost converter to ensure the DC -link voltage V_{dc} to follow closely its reference V_{dcref} (Fig. 19(b)), during the interval time [2.8 s 5 s],

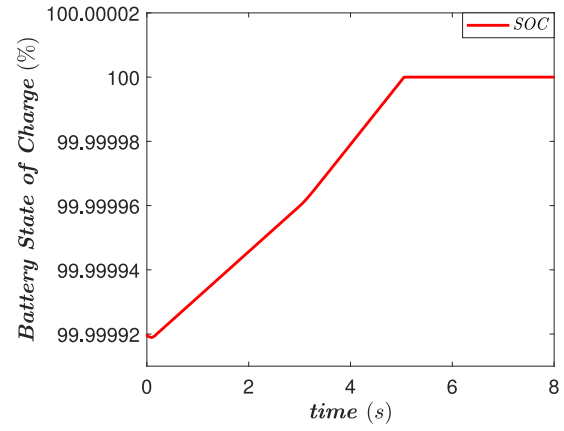
the battery SOC is 20% and the grid is disconnected, in this case, the algorithm management switches to load shedding mode, as shown in Fig. 19(c) and the DC -link voltage V_{dc} remains constant and equal to V_{dcref} , over the interval time [5 s 8 s], the grid is reconnected, since the battery SOC is 20%, the DC -link voltage regulation is ensured via the grid (Fig. 19(b)), as well as the requested load power is delivered by the grid (Fig. 19(c)), the overall control laws are represented in Figs. 19(d)–19(f).

5.4. System's performance in the inrush conditions

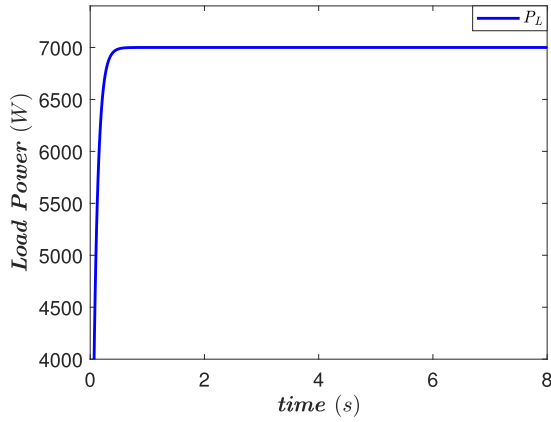
During the testing phase, the system underwent rigorous evaluations under both short and open circuit inrush conditions. The purpose of these tests was to analyze the system's behavior and performance under challenging scenarios. Throughout the experiment, several crucial parameters were closely monitored, including power levels, DC -link voltage, battery current, and battery voltage. These observations were made for two distinct scenarios: the 0.1 s open circuit load case, as illustrated in Fig. 20, and the 0.1 s short-circuit case, depicted in Fig. 21. The gathered results from the comprehensive testing demonstrate the efficacy of the simulation model that incorporates the proposed Back-stepping controllers and power flow management algorithm. The system exhibited correct functionality, effectively handling the diverse and demanding conditions presented during the test. Moreover, the stability of the overall system was confirmed, showcasing its capability to maintain reliable operation even under challenging inrush scenarios. These findings validate the effectiveness and robustness of the implemented



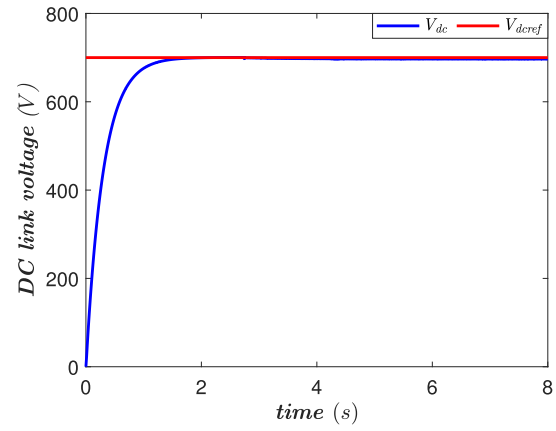
(a) **Top.** The irradiances profile. **Bottom.** The PV power.



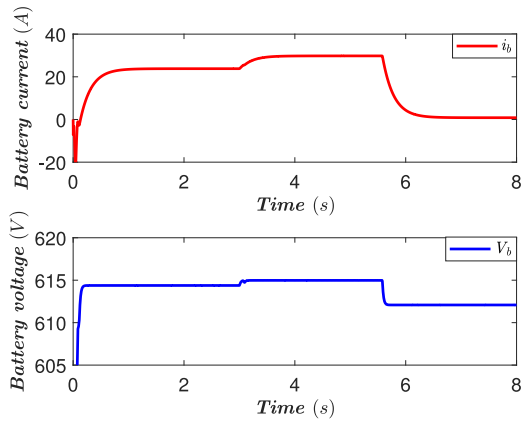
(b) The battery state of charge.



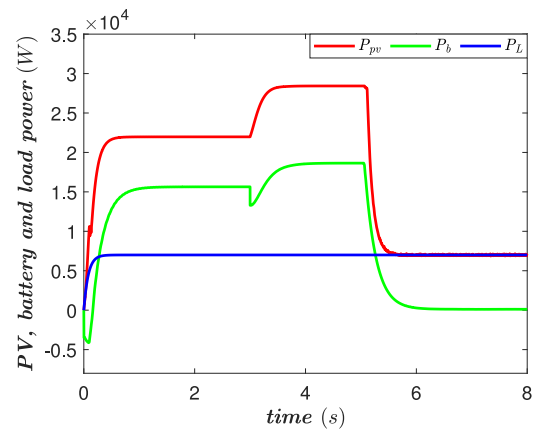
(c) The load power.



(d) **Blue.** The DC-link voltage. **red.** The optimal DC reference voltage.



(e) **Top.** The battery current. **Bottom.** The battery voltage.



(f) **red.** The PV power, **Blue.** The Load power, **Green.** The battery power.

Fig. 14. The power flow management system's performance in the daytime mode with grid OFF. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

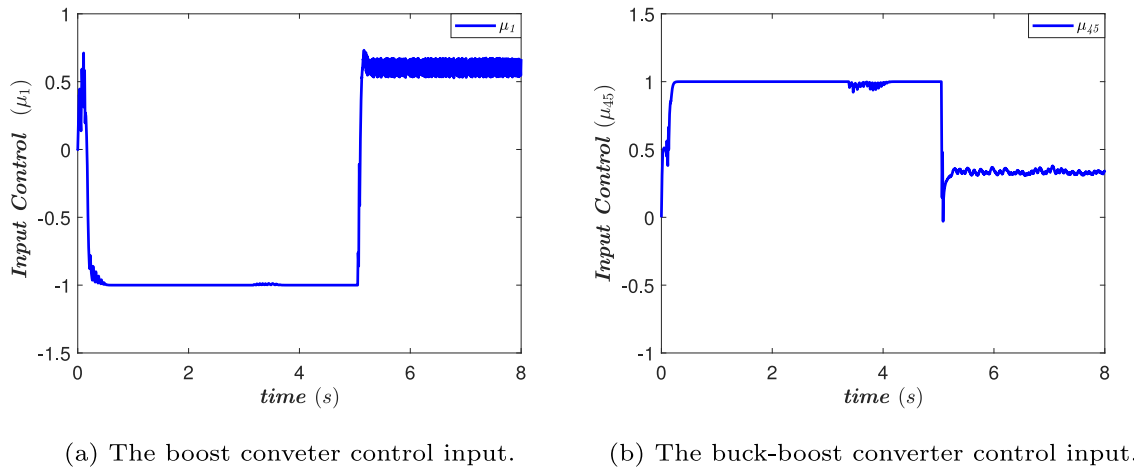


Fig. 15. The average input control laws.

Table 5

Criterion IAE of different modes for $SOC = 100\%$.

		$SOC = 100\%$			
		$P_L = 7 \text{ KW}$		$P_L = 3.5 \text{ KW}$	
		IAE	$R - IAE$	IAE	$R - IAE$
Daytime mode	ON Grid	$0.7547e^{-3}$	100%	$0.6613e^{-3}$	100%
	OFF Grid	$1.0459e^{-3}$	138.58%	$0.9851e^{-3}$	148.96%
Night Mode		$0.9526e^{-3}$	126.14%	$0.7526e^{-3}$	113.80%

Table 6

Criterion IAE of different modes for $SOC = 80\%$.

		$SOC = 80\%$			
		$P_L = 7 \text{ KW}$		$P_L = 3.5 \text{ KW}$	
		IAE	$R - IAE$	IAE	$R - IAE$
Daytime mode	ON Grid	$0.7727e^{-3}$	100%	$0.7425e^{-3}$	100%
	OFF Grid	$1.1054e^{-3}$	143.05%	$0.9931e^{-3}$	133.75%
Night Mode		$0.8516e^{-3}$	110.21%	$0.7906e^{-3}$	106.47%

Table 7

Criterion IAE of different modes for $SOC = 20\%$.

		$SOC = 20\%$			
		$P_L = 7 \text{ KW}$		$P_L = 3.5 \text{ KW}$	
		IAE	$R - IAE$	IAE	$R - IAE$
Daytime mode	ON Grid	$0.7948e^{-3}$	100%	$0.7850e^{-3}$	100%
	OFF Grid	$1.3259e^{-3}$	166.82%	$1.1959e^{-3}$	152.34%
Night Mode		$1.5549e^{-3}$	195.63%	$1.4491e^{-3}$	184.59%

control strategies and power management techniques, underscoring the system's capability to ensure stable and efficient performance in practical applications.

5.5. Comparative study analysis of the integral of absolute error (V_{dc} case)

To further quantify the performance, one calculated the integral of absolute error (IAE) index as follows:

$$IAE_{V_{dc}} = \int |V_{dc} - V_{dc_{ref}}| dt$$

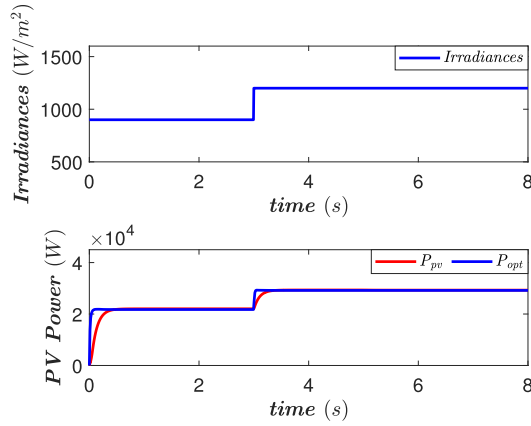
Tables 5, 6, and 7 present the criterion DC-Link IAE (Integral of Absolute Error) for different scenarios under various SOC (State of Charge) and load conditions. These tables showcase the IAE and $R - IAE$ (Relative IAE) values for different modes (Daytime mode with "ON" and "OFF" grid, and Night mode) under different SOC levels (100%, 80%, and 20%) and load conditions (7 kW and 3.5 kW). The

$R - IAE$ values provide a relative comparison with respect to a baseline (e.g., 100%).

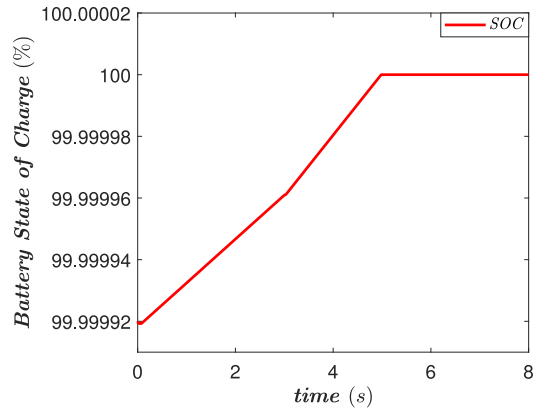
When the system is connected to the grid ("ON" Grid), the IAE values for both load conditions (7 kW and 3.5 kW) and SOC levels (100%, 80%, and 20%) are relatively low, indicating good performance.

On the other hand, when the system is disconnected from the grid ("OFF" Grid), the IAE values are slightly higher for both load conditions and SOC levels compared to the "ON" Grid case. This indicates that the system's performance is slightly compromised when relying solely on local resources. Under the Night mode, the IAE values are generally higher compared to the Daytime mode for both load conditions and SOC levels. This result is a natural consequence, since the AC grid can supply much more energy than the other two sources (Battery and PV).

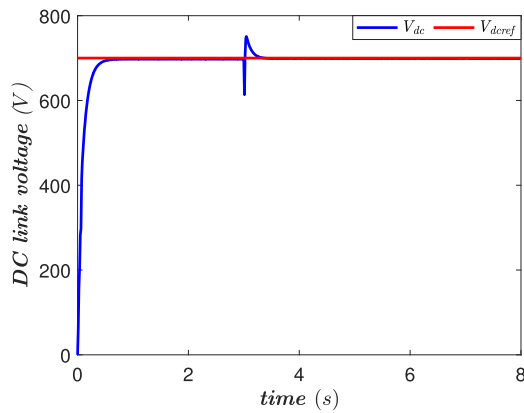
Overall, it shows that the system generally performs well under the Daytime mode when connected to the grid, while the performance is slightly compromised under the "OFF" Grid scenario. The analysis



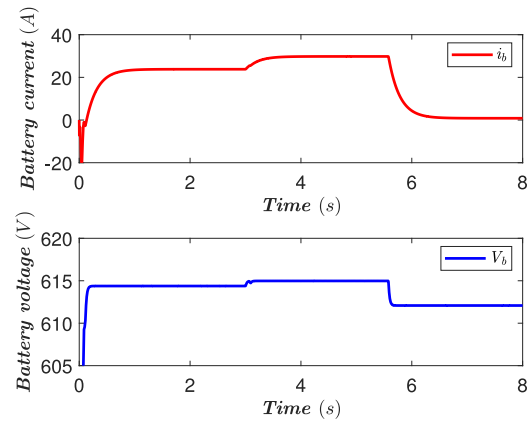
(a) **Top.** The irradiances profile. **Bottom.** The PV power.



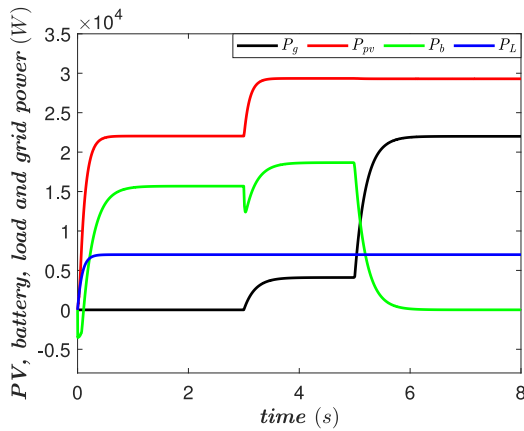
(b) The battery state of charge.



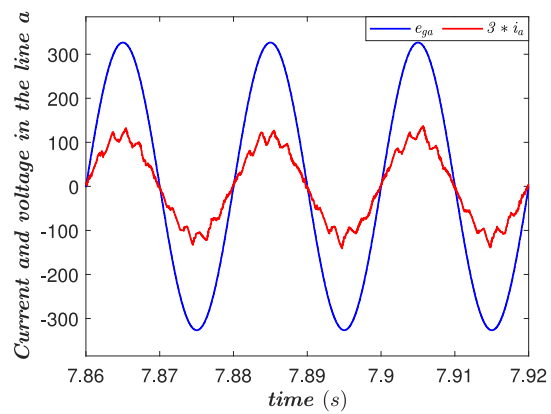
(c) **Blue.** The DC-link voltage. **red.** The optimal DC reference voltage.



(d) **Top.** The battery current. **Bottom.** The battery voltage.

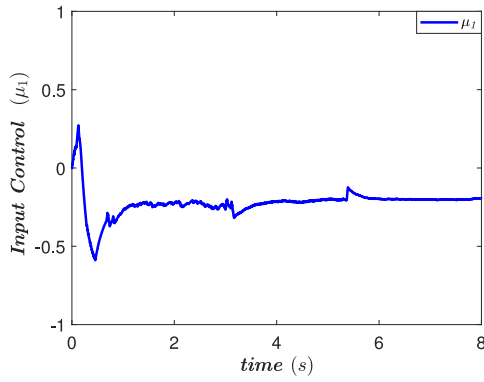


(e) **red.** The PV power, **Blue.** The Load power, **Green.** The battery power, **Black.** The grid power.

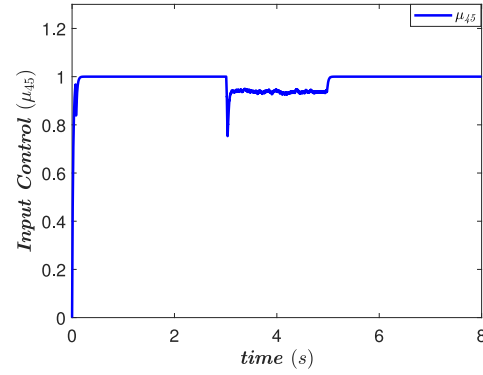


(f) The power factor correction.

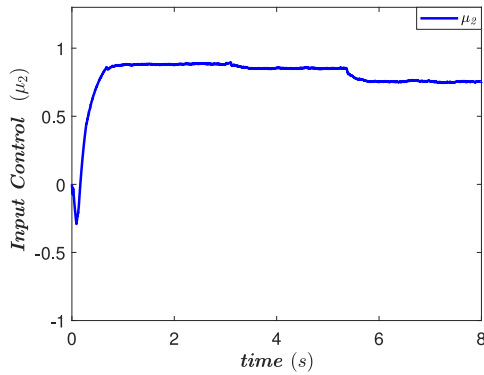
Fig. 16. The power flow management system's performance in the daytime mode with grid ON. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



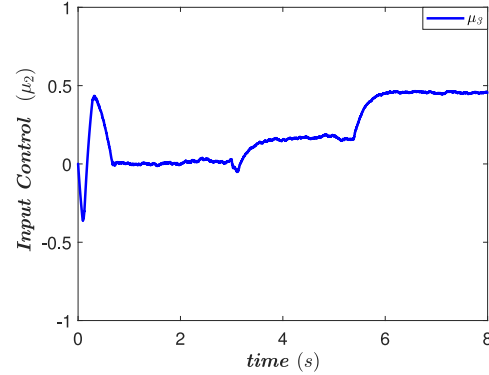
(a) The average boost converter control input.



(b) The average buck-boost converter control input.



(c) The average inverter control input.



(d) The average inverter control input.

Fig. 17. The average input control laws.

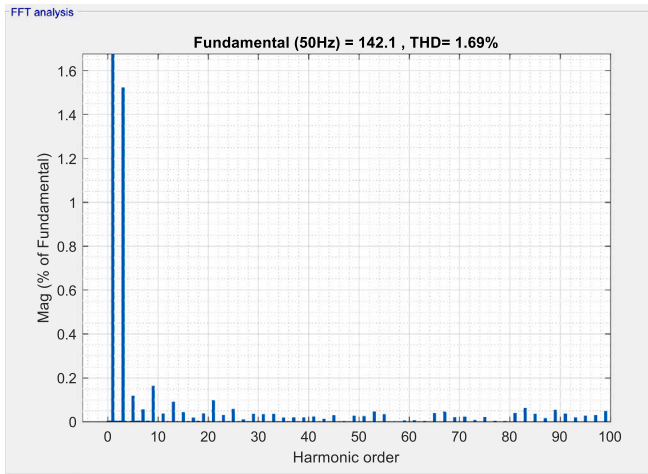
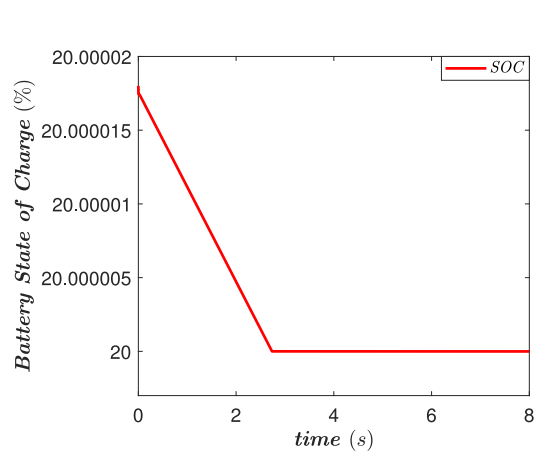


Fig. 18. Grid current i_a spectrum.

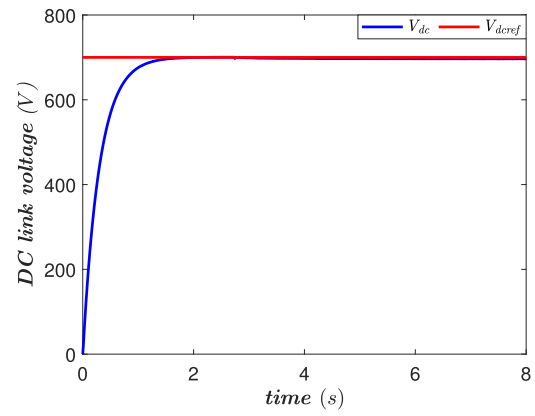
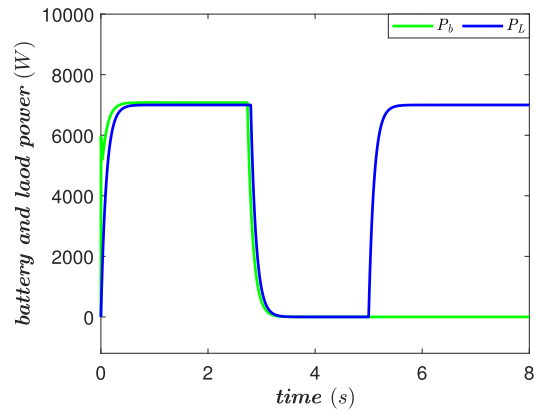
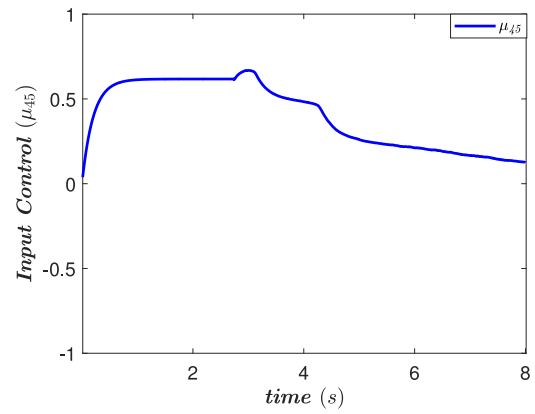
reveals that lower SOC levels have a notable effect on the system's performance, regardless of the operating mode. As the SOC decreases, the errors in achieving the control objectives tend to increase. This highlights the importance of managing and maintaining an adequate SOC level to ensure optimal system performance.

6. Conclusion

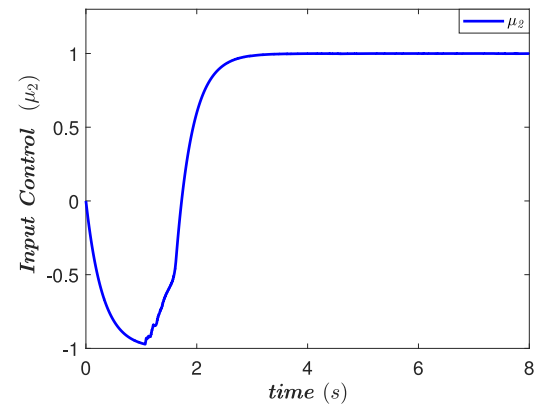
In this study, we have developed a nonlinear control strategy and an energy management algorithm for a solar photovoltaic energy conversion system with an energy storage system. The latter comprises a *PV* generator connected to a three-phase grid through a *DC/DC* boost converter, an inverter, a *DC/DC* buck-boost converter, a lithium-ion battery, and a *DC* load. Firstly, the system dynamics are described by the averaged eighth-order nonlinear state-space model (5a)-(5h). A control strategy is then designed to achieve the following objectives: (i) *PV* voltage regulation: The *PV* panel voltage must track a given reference voltage corresponding to the maximum power point. An artificial neural network (*ANN*)-based optimizer generates the optimal reference voltage using only the current and voltage of the *PV* array as inputs. (ii) *DC*-link voltage regulation: The *DC*-link voltage needs to closely track a reference voltage. (iii) Power factor correction (*PFC*) requirement: The grid currents must be sinusoidal and in phase with the grid voltage. (iv) *CC* mode: When the battery's state of charge is less than 100% ($SOC < 100\%$), indicating that the battery is not fully charged, this mode is adopted. The battery current should closely follow a constant reference within the permissible limits of battery current. (v) *CV* mode: When the battery is completely charged, this mode is activated. The voltage at the battery terminals then reaches its reference signal, corresponding to an *SOC* of 100%. Subsequently, a nonlinear multi-loop control strategy based on Backstepping control is employed to accomplish the control objectives. Formal stability analysis utilizing Lyapunov tools is conducted to ensure system stability. To validate the effectiveness of the proposed control approach, numerical simulations are performed using the



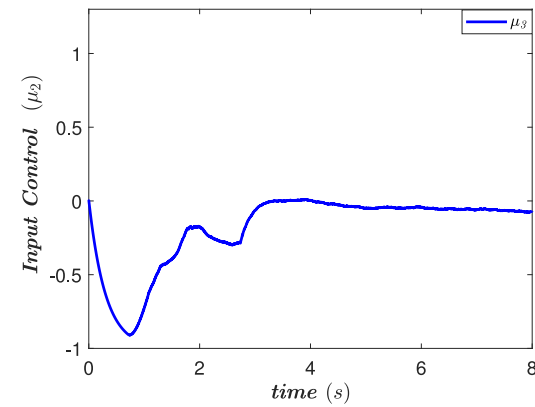
(a) The battery state of charge.

(b) **Blue.** The DC link voltage. **red.** The optimal DC reference voltage.(c) **Blue.** The Load power, **Green.** The battery power.

(d) The average buck-boost converter control input.

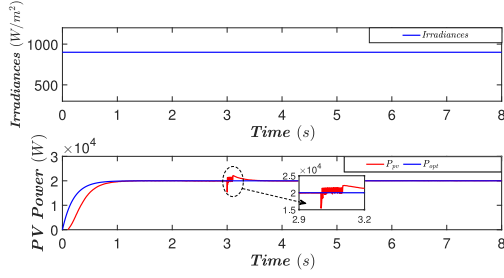


(e) The average inverter control input.

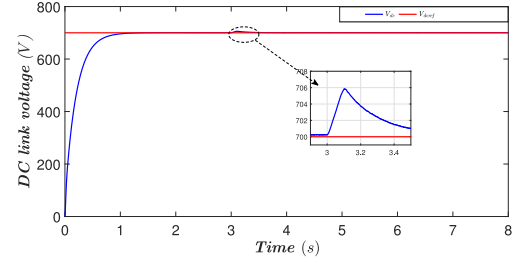


(f) The average inverter control input.

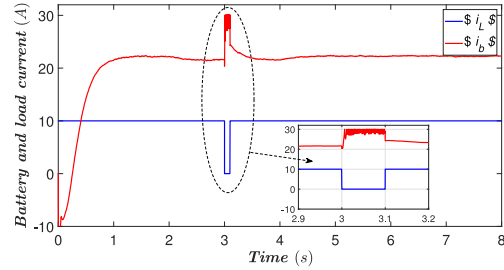
Fig. 19. The power flow management system's performance in the night mode. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



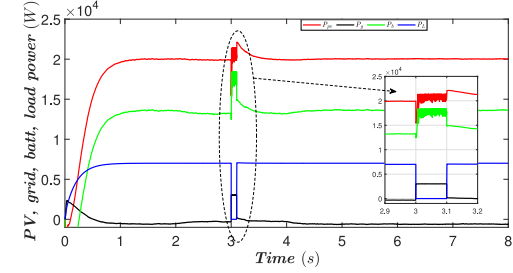
(a) **Top.** The irradiances profile. **Bottom.** The PV power.



(b) **Blue.** The DC-link voltage. **Red.** The optimal DC reference voltage.

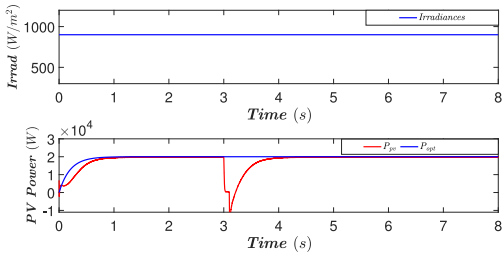


(c) **Red.** The battery current. **Blue.** The load current.

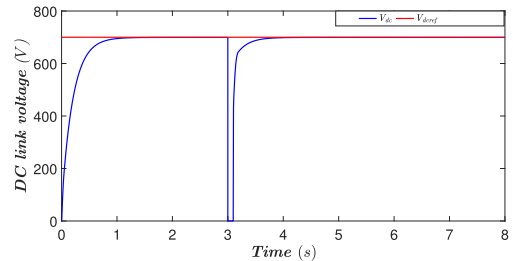


(d) **Red.** The PV power. **Blue.** The Load power. **Green.** The battery power, **Black.** The grid power.

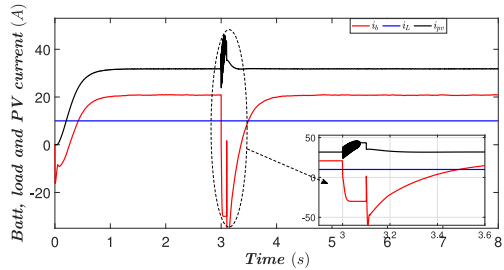
Fig. 20. The power flow management system's performance for load open circuit. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



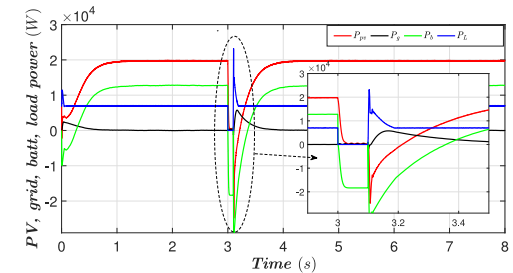
(a) **Top.** The irradiances profile. **Bottom.** The PV power.



(b) **Blue.** The DC-link voltage. **red.** The optimal DC reference voltage.



(c) **Top.** The battery current. **Bottom.** The load current.



(d) **Red.** The PV power, **Blue.** The Load power, **Green.** The battery power, **Black.** The grid power.

Fig. 21. The power flow management system's performance in short circuit case. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

MATLAB/SIMULINK/SIMPOWER environment. The simulation results demonstrate that the control objectives are successfully achieved with excellent performance.

CRedit authorship contribution statement

Karim El Mezdi: Conceptualization, Methodology, Software, Validation, Writing – review & editing. **Abdelmounime El Magri:** Conceptualization, Methodology, Software, Supervision, Writing – review & editing. **Aziz Watil:** Writing – review & editing. **Ilyass El Myasse:** Software, Validation. **Lhoucine Bahatti:** Supervision. **Rachid Lajouad:** Visualization, Investigation, Writing – review & editing. **Hassan Ouabi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix. Proof of propositions and stability analysis

A.1. Proof of Proposition 1

Step 1: Consider the tracking error z_1 :

$$z_1 = x_1 - x_1^* \quad (\text{A.1})$$

Using the Eqs. (8a) and (A.1), the dynamic of the error z_1 undergoes the following equation:

$$\dot{z}_1 = \frac{1}{C_i}(i_{pv} - x_2) - \dot{x}_1^* \quad (\text{A.2})$$

To obtain a stabilizing input for the subsystem (8a), one considers the quadratic Lyapunov function (LF):

$$V_1 = 0.5z_1^2 \quad (\text{A.3})$$

It is easy to verify that $\dot{V}_1$ is a negative definite function of z_1 if the virtual input x_2 is chosen as follows:

$$x_2^* = i_{pv} - C_i[-c_1 z_1 + \dot{x}_1^*] \quad (\text{A.4})$$

Where c_1 is any positive design parameter. If the virtual input x_2 tends towards its reference x_2^* , then $\dot{z}_1 = -c_1 z_1$.

Step 2: The aim, now is to determine the control law. For that, one considers the tracking error z_2 :

$$z_2 = x_2 - x_2^* \quad (\text{A.5})$$

It follows from (A.5) that:

$$\dot{z}_2 = \dot{x}_2 - \dot{x}_2^* = \frac{x_1}{L_i} - \frac{(1-\mu_1)}{L_i} \sqrt{x_3} - \dot{x}_2^* \quad (\text{A.6})$$

To stabilize the subsystem (8b), one can consider the following quadratic Lyapunov function (LF):

$$V_2 = 0.5z_2^2 \quad (\text{A.7})$$

It is easy to check that $\dot{V}_2$ can be made negative definite in the state z_2 by getting:

$$\frac{x_1}{L_i} - \frac{(1-\mu_1)}{L_i} \sqrt{x_3} - \dot{x}_2^* = -c_2 z_2 \quad (\text{A.8})$$

Where c_2 is any positive design parameter. Then, the input control law will be as follows:

$$\mu_1 = 1 - \frac{L_i}{\sqrt{x_3}} \left[\frac{x_1}{L_i} + c_2 z_2 - \dot{x}_2^* \right] \quad (\text{A.9})$$

A.2. Proof of Proposition 2

Step 1: Consider the tracking error z_3 :

$$z_3 = x_3 - x_3^* \quad (\text{A.10})$$

In view of (8c), the dynamic of z_3 undergoes the following differential equation:

$$\dot{z}_3 = -\frac{E_{gd}}{C_{dc}} x_4 - \frac{E_{gq}}{C_{dc}} x_5 + (1-\mu_1) \frac{x_2 \sqrt{x_3}}{C_{dc}} - \frac{(P_b + P_L)}{C_{dc}} - \dot{x}_3^* \quad (\text{A.11})$$

Now, consider the LF candidate $V_3 = 0.5z_3^2$, the time derivative of V_3 can easily be written as:

$$\dot{V}_3 = z_3 \left(-\frac{E_{gd}}{C_{dc}} x_4 - \frac{E_{gq}}{C_{dc}} x_5 + (1-\mu_1) \frac{x_2 \sqrt{x_3}}{C_{dc}} - \frac{(P_b + P_L)}{C_{dc}} - \dot{x}_3^* \right) \quad (\text{A.12})$$

Let the quantity $\theta = -\frac{E_{gd}}{C_{dc}} x_4 - \frac{E_{gq}}{C_{dc}} x_5$ stand as a virtual control signal for (A.11). To ensure that the time derivative of LF will be a negative definite function, the stabilizing virtual control (A.13) is proposed.

$$\theta^* = -c_3 z_3 - (1-\mu_1) \frac{x_2 \sqrt{x_3}}{C_{dc}} + \frac{(P_b + P_L)}{C_{dc}} + \dot{x}_3^* \quad (\text{A.13})$$

As θ is a virtual control input, a new error is introduced.

$$z_4 = \theta - \theta^* \quad (\text{A.14})$$

Then, using (A.11) and (A.13), the dynamic of the error z_3 will now be described by:

$$\dot{z}_3 = -c_3 z_3 + z_4 \quad (\text{A.15})$$

In view of (A.13), it turns out that the z_4 dynamic can be expressed as:

$$\dot{z}_4 = \dot{\theta} - \dot{\theta}^* = -c_3^2 z_3 + c_3 z_4 - \frac{\sqrt{x_3}}{L_g C_{dc}} (E_{gd} \mu_2 + E_{gq} \mu_3) + g(x, t) \quad (\text{A.16})$$

With,

$$g(x, t) = -\frac{E_{gd}}{C_{dc}} \left(-\frac{R_g}{L_g} x_4 + \omega_g x_5 - \frac{E_{gd}}{L_g} \right) - \frac{E_{gq}}{C_{dc}} \left(-\frac{R_g}{L_g} x_5 - \omega_g x_4 - \frac{E_{gq}}{L_g} \right) + (1-\mu_1) \frac{x_2 \sqrt{x_3}}{C_{dc}} + (1-\mu_1) \frac{\dot{x}_2 \sqrt{x_3}}{C_{dc}} - \dot{\mu}_1 \frac{x_2 \sqrt{x_3}}{C_{dc}} - \ddot{x}_3^* \quad (\text{A.17})$$

Consider the following candidate LF:

$$V_4 = 0.5z_3^2 + 0.5z_4^2 \quad (\text{A.18})$$

Using (A.15), the time derivative of (A.18) along the trajectory of (z_3, z_4) gives:

$$\dot{V}_4 = z_3 \dot{z}_3 + z_4 \dot{z}_4 = z_3(-c_3 z_3 + z_4) + z_4 \dot{z}_4 = -c_3^2 z_3 + z_4(\dot{z}_4 + z_3) \quad (\text{A.19})$$

It follows from (A.19), that the LF can be a negative definite function by choosing the z_4 dynamic as follows:

$$\dot{z}_4 = -c_4 z_4 - z_3 \quad (\text{A.20})$$

Combining (A.16) with (A.20), one gets the following stabilizing control laws:

$$\frac{\sqrt{x_3}}{L_g C_{dc}} (E_{gd} \mu_2 + E_{gq} \mu_3) = -(c_3^2 - 1) z_3 + (c_3 + c_4) z_4 + g(x, t) \quad (\text{A.21})$$

Step 2: Now, let us focus on the second control objective, for which a new tracking error is considered as $z_5 = Q_g - Q_{gref}$. In view of (8d) and (8e), the time derivative of the above error is described by the following differential equation:

$$\dot{z}_5 = \frac{\sqrt{x_3}}{L_g} (E_{gd} \mu_2 - E_{gq} \mu_3) + h(x, t) \quad (\text{A.22})$$

With,

$$h(x, t) = E_{gq} \left(-\frac{R_g}{L_g} x_4 + \omega_g x_5 - \frac{E_{gd}}{L_g} \right) - E_{gd} \left(-\frac{R_g}{L_g} x_5 - \omega_g x_4 - \frac{E_{gq}}{L_g} \right) - \dot{Q}_{gref} \quad (\text{A.23})$$

Using the expressions (A.15) and (A.20), the time derivative of $V_5 = 0.5z_3^2 + 0.5z_4^2 + 0.5z_5^2$ is given by:

$$\dot{V}_5 = -c_3 z_3^2 - c_4 z_4^2 - c_5 z_5^2 \quad (\text{A.24})$$

The above expression of V_5 can be made a negative definite function of (z_3, z_4, z_5) using the following control laws:

$$\frac{\sqrt{x_3}}{L_g} (E_{gq} \mu_2 - E_{gd} \mu_3) = -c_5 z_5 - h(x, t) \quad (\text{A.25})$$

Solving the resulting second-order linear equation system (A.21)–(A.25) gives the control laws:

$$\begin{bmatrix} \mu_2 \\ \mu_3 \end{bmatrix} = \frac{L_g C_{dc}}{\sqrt{x_3} E_g^2} \begin{bmatrix} E_{gd} & \frac{E_{gq}}{C_{dc}} \\ E_{gq} & -\frac{E_{gd}}{C_{dc}} \end{bmatrix}^{-1} \begin{bmatrix} -(c_3^2 - 1)z_3 + (c_3 + c_4)z_4 + g(x, t) \\ -c_5 z_5 - h(x, t) \end{bmatrix} \quad (\text{A.26})$$

A.3. Proof of Proposition 3

Let us introduce the battery current tracking error:

$$z_6 = i_b - i_{bref} = \frac{x_7 - x_8 - U_{ocv}}{R_s} - i_{bref} \quad (\text{A.27})$$

In view of (A.27), the dynamic of the above error undergoes the following differential equation:

$$\dot{z}_6 = ax_6 - \frac{a}{R_s} x_7 + \frac{a}{R_s} x_8 + \frac{a}{R_s} U_{ocv} - \frac{b}{R_s} x_7 + \frac{c}{R_s} x_8 + \frac{b}{R_s} U_{ocv} \quad (\text{A.28})$$

The quantity “ $\alpha = ax_6$ ” stands up as a virtual input for the z_6 -dynamic. Consider the quadratic LF candidate: $V_6 = 0.5z_6^2$, the time derivative of V_6 is given by:

$$\dot{V}_6 = z_6 \dot{z}_6 = z_6 \left(ax_6 - \frac{(a+b)}{R_s} x_7 + \frac{(a+c)}{R_s} x_8 + \frac{(a+b)}{R_s} U_{ocv} \right) \quad (\text{A.29})$$

Then one can consider (A.30) as an input control of the sub-system (A.27)

$$\alpha^* = -c_6 z_6 + \frac{(a+b)}{R_s} x_7 - \frac{(a+c)}{R_s} x_8 - \frac{(a+b)}{R_s} U_{ocv} \quad (\text{A.30})$$

Where c_6 is a design parameter. Because “ ax_6 ” is not the real control signal, the following error occurs:

$$z_7 = ax_6 - \alpha^* \quad (\text{A.31})$$

In view of (A.28) and (A.30):

$$\dot{z}_6 = -c_6 z_6 + z_7 \quad (\text{A.32})$$

Stabilizing the system (z_6, z_7) : The primary objective here is to stabilize the state error system (z_6, z_7) by selecting an appropriate control signal “ μ_{45} ”.

$$\begin{aligned} \dot{z}_7 = \mu_{45cc} \frac{a}{L_1} \sqrt{x_3} - c_6^2 z_6 + c_6 z_7 - \left[\frac{ar_1}{L_1} + a(a+b) \right] x_6 \\ - \left[\frac{a}{L_1} - \frac{a(a+b)+b(a+c)}{R_s} \right] x_7 - \left[\frac{a(a+b)+c(a+c)}{R_s} \right] x_8 \\ - \left[\frac{a(a+b)+b(a+c)}{R_s} \right] U_{ocv} \end{aligned} \quad (\text{A.33})$$

To analyze the stability of the error system (z_6, z_7) , let us consider the augmented LF candidate $V_7 = V_6 + 0.5z_7^2$. It follows that V_7 can meet a negative definite function by choosing the z_7 dynamic as follows:

$$\dot{z}_7 = -c_7 z_7 - z_6 \quad (\text{A.34})$$

Finally, from (A.33) and using (A.34), the stabilizing control law μ_{45} of the DC/DC power converter during CC mode is obtained:

$$\begin{aligned} \mu_{45cc} = \frac{L_1}{a\sqrt{x_3}} \left[(c_6^2 - 1)z_6 - (c_6 + c_7)z_7 + \left[\frac{ar_1}{L_1} + a(a+b) \right] x_6 \right. \\ \left. + \left[\frac{a}{L_1} - \frac{a(a+b)+b(a+c)}{R_s} \right] x_7 + \left[\frac{a(a+b)+c(a+c)}{R_s} \right] x_8 \right. \\ \left. + \left[\frac{a(a+b)+b(a+c)}{R_s} \right] U_{ocv} \right] \end{aligned} \quad (\text{A.35})$$

A.4. Proof of Proposition 4

Let us introduce the voltage tracking error as follows:

$$z_8 = x_7 - V_{bref} \quad (\text{A.36})$$

Considering that $\dot{V}_{bref} = 0$ and using (8g), the time derivative of (A.36) yields:

$$\dot{z}_8 = \frac{x_6}{C_1} - ax_7 + ax_8 + aU_{ocv} \quad (\text{A.37})$$

Let the quantity “ $\gamma = \frac{x_6}{C_1}$ ” stand as a virtual control input, and consider the LF candidate: $V_8 = 0.5z_8^2$, the time derivative of the LF (V_8) is given by:

$$\dot{V}_8 = z_8 \left(\frac{x_6}{C_1} - ax_7 + ax_8 + aU_{ocv} \right) = -c_8 z_8^2 \quad (\text{A.38})$$

Which gives the following virtual input law.

$$\gamma^* = -c_8 z_8 + ax_7 - ax_8 - aU_{ocv} \quad (\text{A.39})$$

Where $c_8 > 0$ is a design parameter. Let introduce the new tracking error $z_9 = \frac{x_6}{C_1} - \gamma^*$. In view of (A.37) and (A.39), it follows that the z_8 -dynamic can be described by:

$$\dot{z}_8 = -c_8 z_8 + z_9 \quad (\text{A.40})$$

Now, the goal is to enforce the errors z_9 to converge exponentially to zero. Therefore, one needs the dynamic of the error $z_9 = \frac{x_6}{C_1} - \gamma^*$. Using (8f) and (A.39), one obtains:

$$\dot{z}_9 = \frac{\dot{x}_6}{C_1} - \dot{\gamma}^* \quad (\text{A.41})$$

Which gives:

$$\begin{aligned} \dot{z}_9 = \mu_{45cv} \frac{\sqrt{x_3}}{L_1 C_1} - c_8^2 z_8 + c_8 z_9 - \left(\frac{a}{C_1} + \frac{r_1}{L_1 C_1} \right) x_6 \\ - \left(\frac{1}{L_1 C_1} - a(a+b) \right) x_7 - a(a+c) x_8 - a(a+b) U_{ocv} \end{aligned} \quad (\text{A.42})$$

To establish a stabilizing control law, consider the new quadratic LF candidate:

$$V_9 = V_8 + 0.5z_9^2 \quad (\text{A.43})$$

Which can be derived with respect to time.

$$\dot{V}_9 = -c_8 z_8^2 + z_9 (\dot{z}_9 + z_8) \quad (\text{A.44})$$

In order to make V_9 a negative definite function, the z_9 -dynamics can be defined by:

$$\dot{z}_9 = -c_9 z_9 - z_8 \quad (\text{A.45})$$

Where $c_9 > 0$ is a new design parameter. Finally, from (A.42) and using (A.45), the control law μ_{45} of the DC/DC power converter when CV mode control is required can be obtained as follows:

$$\begin{aligned} \mu_{45cv} = \frac{L_1 C_1}{\sqrt{x_3}} \left[(c_8^2 - 1)z_8 - (c_8 + c_9)z_9 + \left(\frac{a}{C_1} + \frac{r_1}{L_1 C_1} \right) x_6 + \left(\frac{1}{L_1 C_1} - a(a+b) \right) x_7 \right. \\ \left. + a(a+c) x_8 + a(a+b) U_{ocv} \right] \end{aligned} \quad (\text{A.46})$$

A.5. Proof of Proposition 5

Step 1: Consider the tracking error $z_{10} = x_3 - x_3^*$, and using the subsystem (8c) and Eq. (17), the dynamic of error z_{10} undergoes the following equation:

$$\dot{z}_{10} = -\frac{P_b}{C_{dc}} - \frac{P_L}{C_{dc}} - \dot{x}_3^* \quad (\text{A.47})$$

To obtain a stabilizing input for the subsystem (A.47), one considers the quadratic LF : $V_{10} = 0.5z_{10}^2$. It is easy to verify that $\dot{V}_{10}$ is a negative definite function of z_{10} if the virtual input P_b is chosen as follows:

$$P_b^* = C_{dc}[c_{10}z_{10} - \frac{P_L}{C_{dc}} - \dot{x}_3^*] \quad (\text{A.48})$$

Where c_{10} is any positive design parameter. If the virtual input P_b tends towards its reference P_b^* , then $\dot{z}_{10} = -c_{10}z_{10}$.

Step 2: The aim, now is to determine the control law. For that, one considers the tracking error $z_{11} = P_b - P_b^*$. It follows:

$$\dot{z}_{11} = \dot{P}_b - \dot{P}_b^* = -(\frac{r_1}{L_1} + a)P_b + \mu_{45}\frac{x_7\sqrt{x_3}}{L_1} + f(x, t) - \dot{P}_b^* \quad (\text{A.49})$$

To stabilize the subsystem (A.49), one can consider the quadratic LF as: $V_{11} = 0.5z_{11}^2$. Then, it is easy to check that $\dot{V}_{11}$ can be made negative definite in the state z_{11} by getting:

$$-(\frac{r_1}{L_1} + a)P_b + \mu_{45}\frac{x_7\sqrt{x_3}}{L_1} + f(x, t) - \dot{P}_b^* = -c_{11}z_{11} \quad (\text{A.50})$$

Where c_{11} is any positive design parameter. Then, the input control law will be as follows:

$$\mu_{45dc} = \frac{L_1}{x_7\sqrt{x_3}}[-c_{11}z_{11} + (\frac{r_1}{L_1} + a)P_b - f(x, t) + \dot{P}_b^*] \quad (\text{A.51})$$

A.6. Proof of Proposition 6

Considering the following tracking error $z_{12} = x_3 - x_3^*$, and using the Eq. (8c), the dynamic of error z_{12} undergoes the following equation:

$$\dot{z}_{12} = (1 - \mu_1)\frac{x_2\sqrt{x_3}}{C_{dc}} - \frac{E_{gd}}{C_{dc}}x_4 - \frac{E_{gq}}{C_{dc}}x_5 - \frac{(P_b + P_L)}{C_{dc}} - \dot{x}_3^* \quad (\text{A.52})$$

To obtain a stabilizing input for the subsystem (A.52), one considers the quadratic LF as: $V_{12} = 0.5z_{12}^2$. It is easy to check that $\dot{V}_{12}$ can be made negative definite in the state z_{12} by getting:

$$(1 - \mu_1)\frac{x_2\sqrt{x_3}}{C_{dc}} - \frac{E_{gd}}{C_{dc}}x_4 - \frac{E_{gq}}{C_{dc}}x_5 - \frac{(P_b + P_L)}{C_{dc}} - \dot{x}_3^* = -c_{12}z_{12} \quad (\text{A.53})$$

Where c_{12} is any positive design parameter. Then, the input control law will be as follows:

$$\mu_{1b} = 1 - \frac{C_{dc}}{x_2\sqrt{x_3}}\left[-c_{12}z_{12} + \frac{E_{gd}}{C_{dc}}x_4 + \frac{E_{gq}}{C_{dc}}x_5 + \frac{(P_b + P_L)}{C_{dc}} + \dot{x}_3^*\right] \quad (\text{A.54})$$

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